Grammaticality de-idealized

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Abstract

This paper introduces the Operator-Value Model of Grammaticality (OVMG), a framework for (un)grammaticality distinct from grammars focused solely on language production or description. The model treats (un)grammaticality as conditioned stability in form–value relations accepted or rejected within communicative situations, integrating mapping viability, operator-value coherence, population licensing, and speaker-level feelings of anomaly. Previous approaches, including generative grammar, Construction Grammar, and psycholinguistic models, have each captured part of the picture but haven't jointly explained why semantically coherent constructions may nonetheless be deemed ungrammatical, why certain transparent structures remain systematically unentrenched, and why categorical grammaticality clusters where it does. The framework derives categoricality from conventionalized norms, opportunity structure, and preemption dynamics within communicative situations, rather than from structural membership or intuition alone. It earns its keep by projection: it predicts which rejected constructions should soften under repeated exposure and which should resist, how learners should treat zero evidence under different opportunity structures, how early second-language rejections should track first-language preemption, and where community treatment of a form should change discontinuously as licensing crosses a situational threshold.

Keywords: grammaticality; projectibility; acceptability; preemption; entrenchment; conventionalization; usage-based grammar

Introduction

Every competent speaker of English knows that **Can the have running* is impossible, but the source of this certainty proves remarkably elusive. What does it mean to say a sentence is ungrammatical? Consider these examples:

(1) a. **Can the have running?*
b. *Colorless green ideas sleep furiously.* (Chomsky, 1957)
c. **I've finished it yesterday.*
d. *?I saw Joan, a friend of whose was visiting.*
e. *The bread the baker the apprentice helped made is delicious.*
f. A: *How old are you?* B: **I have 25 years.*
g. **Which did you buy car?*

While all might receive asterisks in many analyses, they represent fundamentally different types of unacceptability. Some constructions, like (1a), fail to establish any form–value relationship in English. Others, like (1c), involve a clash between the time semantics of clause tense and the word *yesterday*. Still others, such as (1d), show gradient or indeterminate status. (1e) illustrates what might be called theoretically predicted acceptability: constructions that linguistic theory predicts should be grammatical but that are consistently rejected by native speakers in ordinary judgment tasks. Still others, like (1f), are transparent and grammatical in related languages but lack licensing in English age predication. And a few, like the left-branch extraction of (1g), seem to be ruled out entirely, despite being apparently short and interpretable.

Different traditions emphasize different pieces: formal approaches focus on abstract principles, usage-based theories on frequency and entrenchment, processing accounts on cognitive constraints. Each captures part of the picture, but none unifies categorical constraints, gradient acceptability, cross-linguistic variation, and change over time. I'll argue that the obstacle is a shared idealization, and that removing it reorders the questions.

The de-idealization in the title is specific. Early generative grammar idealized grammaticality to the knowledge of an ideal speaker–listener in a completely homogeneous speech community (Chomsky, 1965, p. 3), treating variation, uncertainty, and change as noise around a categorical competence fact. That idealization bought tractability, but it priced out the phenomena in (1). Following Goodman (1955), removing it reorders the questions: ask first what the category GRAMMATICAL lets us predict (its projectibility, what observing some features licenses us to expect about others), and only then what worldly profile supports those predictions and what stabilizes it (Boyd, 1999; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). Mechanisms matter because they explain why the projection holds, not because naming them settles the kind.

This paper develops that answer as the Operator-Value Model of Grammaticality (OVMG). Its unit is the OPERATOR value: a closed-paradigm, update-configuring contrast (tense, agreement, clause type, and their kin), identified by its public-update role rather than by expressive substance, so that tone, particles, word order, and inflection all count when they realize such a contrast (Reynolds, 2026c). The framework rests on three premises, which track the order just named: profile, projection, and support.

1. Profile. Grammaticality is a conditioned form–value relation, specified by whether an operator value is structurally realizable, coherent, and licensed within a communicative situation (community, dialect, register). This profile is disturbed in distinct ways (no licensable analysis, clashing construals, absent situational licensing, processing overload), so the phenomena traditionally lumped as "ungrammatical" in (1) fall into separable modes rather than one kind.
2. Projection. The profile earns its standing by what it lets us predict, not by the mechanisms that hold it together: which rejections should soften under repeated exposure and which resist (1d) vs. (1g), how faithfully forms transmit, and how they change over time.
3. Support, graded. The categorical appearance of grammaticality emerges when usage stabilizes the population's licensing state; the subjective feeling of ungrammaticality is a separate, gradient signal, not grammatical status itself.

Together, the premises explain why the traditions have talked past one another. The profile premise preserves the formal insight that constraints can be systematic while letting categorical and gradient cases share one architecture, in line with Francis's (2022) case for building gradience into core grammatical theory. The projection premise keeps the account testable: low-evidence constructions should respond to exposure and framing differently from high-opportunity preempted forms. The support premise explains why formal licensing, processing cost, and subjective feeling can move together without being the same thing.

Section 1 traces the impasse in grammaticality theory to the attempt to treat it as one phenomenon. Section 2 presents OVMG, defining the three constitutive variables and the recurrent instability modes that route a form to one diagnostic category or another. Section 3 gives the formal core: a state theory that specifies the profile as a population licensing posterior, and a dynamics module that shows how such states arise and stabilize. Together, they derive categoricality from usage rather than stipulating it. Section 5 situates the account against generative grammar, Construction Grammar, usage-based, logicality, and relevance-theoretic approaches, and states the predictions. The paper closes with limitations and a research program spanning corpus, experimental, and cross-linguistic methods.

1 The impasse in grammaticality theory

The concept of grammaticality remains elusive despite its centrality to linguistic theory. After decades of research, the field still lacks a unified account of what makes an utterance grammatical or ungrammatical. The impasse persists because grammaticality is often treated as a single phenomenon, when it actually arises through the interaction of three distinct sources of failure. Understanding why previous approaches have struggled requires disentangling three components: structural viability, interpretive coherence, and situational licensing.

1.1 The problem of structural viability

The first challenge in defining grammaticality is distinguishing structures that admit no viable analysis from those that fail for other reasons. Chomsky (1957), building on formal production systems developed by Post (1943), treated grammaticality as a categorical property defined by membership in a set of well-formed strings. That categorical view captures the intuition that some strings, like (1a), resist structural analysis entirely:

(2) **Can the have running?*

Here, the input crashes the structural analysis independent of meaning. The "categorical" view correctly identifies that a viable structural analysis is necessary for grammaticality. But by raising this single condition to the *definition* of grammaticality, formal approaches struggled with evidence that speakers consistently provide gradient judgments for structures that are clearly analyzable but degraded. The competence-performance distinction (Chomsky, 1965) attempted to preserve categorical grammar by attributing gradience to processing limitations. But as Schütze (2016, p. 71) notes, this often leads to circularity, where problematic results are dismissed as performance artifacts. The OVMG framework resolves this by recognizing that while structural viability (the successful mapping of form to category) is a necessary, categorical prerequisite, it isn't the sole determinant of grammaticality.

1.2 The problem of interpretive coherence

The second tension arises from the interaction of form and meaning. Chomsky's famous example (1b) shows that operator-valued form–value relations can be intact even when compositional semantics fails: the sentence is structurally a declarative, the subject is correctly identified both positionally and through agreement, and the tense suggests a general claim. What breaks is the compositional semantic relation: the lexical values don't cohere. Conversely, (1c) **I've finished it yesterday* shows compositional semantics can be transparent (the intended meaning is clear) while an operator-valued form–value relation fails. Here, the clash is between the present perfect's temporal value and the adverb's deictic anchoring.

Generative semanticists like Lakoff (1971) and McCawley (1968) demonstrated early on that many constraints have semantic motivations. Construction Grammar (Goldberg, 1995) has further shown how constructions carry meanings that interact with lexical semantics. Novel uses that align with established constructional meanings (like *She texted him the address*) are accepted, while those that clash (like **She disappeared him the evidence*) are rejected. A complete theory needs to account for interpretive coherence: the degree to which a structurally viable form yields a stable, non-contradictory interpretation.

1.3 The problem of situational licensing

Finally, even structurally viable and semantically coherent forms may be ungrammatical if they violate community conventions. Sociolinguistic research (Labov, 1972) demonstrates that grammaticality references community norms. Cross-linguistic variation shows that each language community conventionalizes particular form–value mappings through historical processes. These conventions become entrenched through statistical preemption (Goldberg, 2011): speakers learn that certain forms are ungrammatical precisely because they encounter alternative expressions in contexts where the ungrammatical form would otherwise be expected.

Consider (1f), repeated here:

(3) A: *How old are you?* B: **I have 25 years.*

This utterance is structurally viable and semantically transparent (the intended meaning is clear), but it's categorically rejected in English. The same form is grammatical in French (*J'ai 25 ans*) and Spanish (*Tengo 25 años*); the constraint is community-specific. Usage-based approaches (Bybee, 2006) emphasize that such rejection arises from the absence of situational licensing. The English-speaking community has conventionalized a different way to express age, leaving this form without positive evidence.

These three problems (structural viability, interpretive coherence, and situational licensing) correspond exactly to the three constitutive variables of the OVMG framework presented in Section 2. Acknowledging them as distinct but interacting components makes it possible to move beyond the impasse of trying to force all grammaticality judgments into a single "competence" definition or a purely probabilistic usage model.

2 The Operator-Value Model of Grammaticality

The term VALUE is used here in the Saussurean sense of *valeur* (Saussure, 1916): the identity of a linguistic unit is constituted not by intrinsic properties but by its position in a system of contrasts, what it patterns with, what it opposes, and what interpretations it makes available. This isn't feature-value assignment or decision-theoretic value. Throughout, I treat grammatical knowledge as a conditioned form–value relation; when this relation is sufficiently stable in a communicative situation, with a dominant value reliably recoverable and socially licensed, I refer to it informally as an established form–value relationship.

The model's name marks its unit: operator values (closed-paradigm, update-configuring contrasts). A companion operator-stratum paper develops this boundary in full and argues that it follows public-update role rather than expressive substance: tone, particles, word order, inflection, and suprasegmental material all fall within its scope when they realize an operator contrast (Reynolds, 2026c).

Here a self-contained working test suffices, because the concept has to stand on its own inside this paper: it fixes the scope of K, the categoricality prediction, and the appendix on Turkish suffix harmony (§A).

A contrast is an operator contrast when it meets three conditions jointly: drawn from a *closed* paradigm, *high-opportunity* (the communicative job recurs constantly), and *configured to the public update* (clause type, polarity, tense and aspect anchoring, agreement and role allocation, scope, or reference tracking). Mis-setting such a contrast changes the update instruction even when the intended message is recoverable.

These conditions exclude cases that "closed-class", "grammaticalized", or "high-frequency" alone would admit. A closed-class item that doesn't configure the update, such as a purely stance-marking discourse particle, fails the third condition and is predicted *not* to attract categorical policing; a high-frequency open-class collocation fails the first; and a genuinely update-configuring contrast that's gone low-opportunity, such as a moribund case distinction, fails the second and is predicted to destabilize. The prediction is falsifiable in the direction that matters: a closed, high-opportunity contrast that isn't update-configuring should be policed like style, not like grammar.

The formal architecture rests on three constitutive variables: mapping viability (map), operator-value coherence (K), and situational licensing (C_t). Together they specify a PROFILE in the projectibility-first sense: a pattern whose partial observation licenses further expectations (Reynolds, 2026a). Their interaction determines grammaticality (§3); what the profile projects, and what supports those projections, is the business of §3 and §4. Figure 1 lays out this order of explanation, which the rest of the paper fills in. Table 1 summarizes these variables; the formal definitions appear in §3.

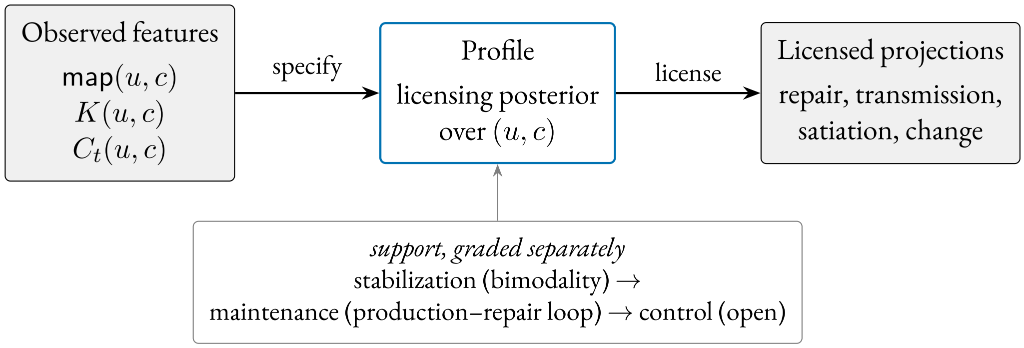

Figure 1: The projectibility-first order of explanation. The three constitutive variables specify a profile (the population licensing posterior) whose partial observation licenses the projections on the right. What stabilizes, maintains, or controls the profile is a separate, graded claim (below), not something inherited from naming the mechanism.

Variable	Component
$\mathsf{map}(u, c)$	Mapping Viability (Categorical prerequisite)
$K(u, c)$	Operator-Value Coherence
$C_t(u, c)$	Situational Licensing (Social convention)

Table 1: The three constitutive variables of the OVMG framework.

The formal core needs names for recurrent INSTABILITY MODES: ways of driving map, K, or C_t toward zero, with processing and prescriptive factors modulating how grammatical something *feels* without directly fixing grammatical status.

2.1 Diagnostic categories

The categories below are labels for common routes to ungrammaticality, not primitives of the model. Each traces to one or more constitutive variables, as the formal core in §3 makes explicit.

FORM–VALUE RELATIONS. Grammaticality depends on stable links between forms and the operator values they realize within communicative situations. That form–value relations exist at every level of grammatical organization, including morphemes and abstract syntactic patterns, is a core insight of Construction Grammar (Goldberg, 1995). OVMG narrows the target of categorical grammaticality to closed-paradigm, high-opportunity, update-critical contrasts: resources that configure how an utterance enters public coordination.

Operator exponents, like words, are typically polysemous: a single form participates in multiple form–value relationships, usually with one dominant value (Kilgarriff, 2004). The English past tense, for instance, usually denotes past time, but it can also mark social distance (*Could you help me?*) or low likelihood (*If I went tomorrow…*). What matters for grammaticality is whether a stable relationship exists within the relevant communicative situation, not whether the form has a single fixed meaning.

Neuroimaging work provides converging evidence for this integration of form and meaning at the processing level. Fedorenko and colleagues find that brain regions activated by syntactic manipulations overlap substantially with those activated by semantic ones (Fedorenko et al., 2011, 2012, 2024). The overlap points to partly shared processing resources rather than wholly separate modules.

Following Wiese (2023, p. 3), grammaticality is conditioned by regularities within a "communicative situation". The emergence claim defended below is narrower: categoricality arises through the licensing dynamics in §4.4, not through an unanalyzed appeal to usage.

Communicative situations aren't static or mutually exclusive. A speaker orients to overlapping contexts that shift on multiple dimensions: the immediate situation (formal meeting vs. casual chat), durable social positions (regional, generational), and the norms they're orienting to (which can shift with audience). This fluidity explains style-shifting, dialect formation, and the gradual acceptance of initially marginal constructions.

Early child language shows this. Toddlers produce utterances that deviate from adult norms but remain internally consistent within the child's developing system. A toddler in a monolingual English household might say:

(4) *Ava cookie.* (intended as 'Ava = I want a cookie')

Although this differs from adult English norms, it may not be perceived as ungrammatical by the child's regular caregivers. Used with the same pragmasemantic force among anglophone adults, the construction would be judged ungrammatical.

Multiple modal constructions provide another clear example of variety-relative grammaticality:

(5) *I might could help you with that.*

This combination of modal auxiliaries is systematically possible for some American English speakers, who can productively generate similar constructions (Morin & Grieve, 2024). Speakers from communities where only single modals are grammatical, though, typically reject such combinations as ungrammatical.

A similar pattern appears in code-mixing among bilingual speakers, where combinations of forms from different languages can be grammatical within that bilingual community's norms but not without. Consider a Spanish-English bilingual speaker who uses a Spanish progressive auxiliary with an English lexical verb:

(6) *Ayer, estábamos lifting en el gym durante una hora.*
yesterday be.IMPF-1PL lifting in the gym for an hour
'Yesterday, we were lifting in the gym for an hour.'

Within the right communicative situation, this utterance is grammatical. The Spanish auxiliary *estábamos* combines with an English participial form *lifting* to form the progressive aspect. This cross-linguistic combination of morphology and a lexical verb is consistent with local norms, where code-mixed utterances are common and meaningful. In contrast, in a standard monolingual Spanish setting, where fully Spanish progressive structures are licensed (*estábamos levantando pesas*), speakers may judge the example in (6) as ungrammatical. The use of intransitive *lifting*, specific to the gym community, further illustrates just how localized grammaticality judgments can be.

This doesn't mean bilingual communities simply accept any combination of languages. As Toribio (2001) reports, Spanish–English bilingual speakers judge examples like (7) as unacceptable, showing that even in bilingual communities, there are systematic constraints on which language combinations are permitted.

(7) * *Los enanitos intentaron pero no succeeded in awakening Snow White* (Toribio, 2001)
the dwarfs try.PST-3PL but not succeeded in awakening Snow White
'The dwarfs tried but didn't succeed in awakening Snow White.'

In a slightly different case, as a second-language speaker of Japanese, I used to say

(8) * *Kawaii da.*
cute-PRES COP.PRES
'(That)'s cute.' (intended)

Conventionally, though, the redundant tense marking has no accepted meaning, making the use of the copula ungrammatical to most Japanese speakers, though it felt meaningful and grammatical to me. The example shows why acquisition history matters: shared routines stabilize form–value relations for a community, while individual trajectories can still pull judgments away from that norm.

The specific linguistic context can matter too:

> To take an obvious case which Jerry Morgan [(1973)] discussed recently, certain combinations of words are extremely strange if presented in isolation but are perfectly normal as answers to certain questions. *Spiro conjectures Ex-Lax* would generally be felt to be unintelligible if presented out of context but is a perfectly normal answer to the question *Does anyone know what Mrs. Nixon frosts her cakes with?* (McCawley, 1974, 252, italics added)[1]

These examples show why grammaticality has to be indexed to communicative situation. Conversants who differ in immediate situation, social position, or the norms they orient to may disagree on whether a form is grammatical, and both may be right within their respective contexts. The linguist's task is to determine whether a form is grammatical from some position in that space, or systematically excluded from all of it.

The stability of form–value relations within speech communities can be empirically modeled, as demonstrated by Blythe and Croft (2009). Their utterance selection model treats speech communities as networks where speakers track and reproduce linguistic variants based on their interactions. When applied to dialect formation, these models show how competing linguistic variants spread and eventually stabilize, with initial variant frequency strongly predicting which form will prevail.

In different languages, different distinctions become conventionalized as grammatically obligatory. These obligatory contrasts reflect historically stabilized patterns in which meaning distinctions are systematically marked. Over time, usage establishes grammatical constructions that reliably signal these distinctions. As a result, what counts as a grammatical necessity in one language may be optional or absent in another.

The progressive aspect provides a useful example. In Standard English, it isn't optional but an obligatory grammatical marker for ongoing, incomplete actions:

(9) *She is studying right now.*

[1] The humour derives from political tensions of the Watergate era, when both Vice-President Spiro Agnew and President Nixon would ultimately resign from office.

Here, the progressive form *is studying* isn't just a stylistic choice. It's the recognized, grammatical way to convey the appropriate aspectual meaning in this frame.

In French, by contrast, the progressive aspect isn't a grammatically mandatory distinction. Speakers can signal ongoing activity through adverbs or periphrastic constructions, but standard French doesn't have a dedicated progressive form. The sentence:

(10) *Elle étudie maintenant.*
'she studies/is studying now'

can comfortably describe a currently ongoing action without any need for special morphology. In Standard French, this aspectual distinction isn't conventionalized as an obligatory operator contrast. English grammar treats the contrast as obligatory; French doesn't.

The same holds for evidentiality, the grammatical marking of information sources. In Turkish, evidentiality is systematically realized through verb forms and particles that distinguish between directly witnessed events and those inferred or reported:

(11) *Gelmiş*
'He/she came (apparently)'
(i.e. the speaker wasn't a witness but inferred or heard about it.)

In Turkish, evidential distinctions are conventionalized as obligatory operator contrasts. A speaker can't simply omit evidential marking without sounding ungrammatical.

English speakers can say *I heard that he arrived* or *He must have arrived*, but these are optional lexical or modal resources rather than required elements of the grammar. In English, evidential distinctions aren't conventionalized as obligatory operator contrasts.

Kilani-Schoch and Dressler (2005) demonstrate this principle systematically in their analysis of verbal morphology across French and other Romance languages. They show how apparently similar verbal systems can grammaticalize quite different semantic distinctions as obligatory. The contrast reflects different conventions about which meaning distinctions are systematically marked. For instance, while both French and Italian mark aspect morphologically, they differ in which aspectual distinctions are grammaticalized versus left to optional lexical expression.

Establishing a form–value mapping is necessary for grammaticality, but it isn't sufficient. Speakers routinely interpret strings that still count as ill-formed in the relevant communicative situation. That flexibility makes licensing, not interpretability in principle, the central question.

Three further pressures sort candidate mappings: conflict with conventional constructional values contributes negative evidence; high recovery cost down-weights the mapping; and weak conventionalization leaves the mapping with insufficient positive evidence. Their interaction predicts why **Furiously sleep ideas green colorless* elicits universal rejection while (1e) feels bad in everyday quotation but is accepted as "grammatical but hard to process" once speakers are walked through the intended structure. The weighting concentrates probability mass on the smaller subset of imaginable relations recognized as bona fide grammatical resources.

Nor does convergent usage by itself settle the matter. Defining grammaticality as "whatever speakers accept" would be circular: it couldn't flag contested innovations, explain stable dialectal differences despite mutual intelligibility, or distinguish licensed forms from performance errors that pass unnoticed. Separating mapping, compatibility, and licensing avoids trivializing grammaticality while still treating durable population-level licensing as decisive for which relationships endure.

Operator-value compatibility. Beyond the existence of form–value relations, grammaticality requires compatibility among the operator values those relations realize. This component captures cases where individual elements are well formed but their combination creates conflicts among grammaticalized contrasts activated in the communicative situation.

Consider the temporal incompatibility in:

(12) * *I've finished it yesterday.*

Here, the present perfect construction realizes a current-relevance operator value while *yesterday* specifies completed past time. Unlike lexically incongruous combinations like *colorless green ideas*, which remain grammatical because the clash doesn't mis-set an operator value, this example shows a direct conflict between the temporal value realized by the grammatical construction and the lexical temporal specification.

The age example belongs elsewhere:

(13) * *I have 25 years.* intended as 'I'm 25 years old'

In English, *have+years* is a licensed frame for duration, remaining time, or experience, but ordinary age predication is licensed through a different frame. The intended value is transparent, so the failure isn't a K failure in the narrow sense. It's a licensing failure for an age-predication construction in English, even though cognate-looking frames are licensed in French and Spanish.

Sometimes value incompatibility involves conflicting information-structural requirements:

(14) * *Who did the lifeguard who saved _ work in New Jersey?* (Cuneo & Goldberg, 2023)

The clash is in information structure: the same participant is simultaneously focused through fronting *who* while being backgrounded by the relative clause construction (Cuneo & Goldberg, 2023).

These examples also delimit the coherence term K. K measures whether the best available construal satisfies the grammaticalized operator constraints that are active in the communicative situation. In the energy notation introduced below, those active constraints are the ϕ_j terms: tense–aspect anchoring where tense–aspect is obligatory, argument-linking requirements where a construction licenses roles, information-structural requirements where a construction grammatically packages focus or backgrounding, and indexical contrasts only when they've entered a closed accountable repertoire.

This makes K partly derivative of licensing facts at the feature level, keeping arbitrary lexical implausibility out of grammatical incoherence.

Processing constraints. Language processing engages both dedicated language networks and domain-general cognitive systems (Fedorenko et al., 2024). When constructions overload these systems, particularly through multiple long-distance dependencies or heavy embedding, they may trigger feelings of ungrammaticality despite being structurally well-formed.

(15) *The bread the baker the apprentice helped made is delicious.*

Evidence for these constraints comes from multiple sources. Studies of sentence processing show that dependencies spanning multiple intervening elements increase processing difficulty, with new referents between dependent elements compounding memory load (Gibson, 2000, 2026). In (15), while each individual relation (like *the apprentice helped* and *the baker made*) is interpretable in isolation, their nested combination overwhelms incremental processing. The construction is grammatical, but excessive bridging costs from its nested structure trigger feelings of ungrammaticality.

Rather than reflecting simple memory limitations, processing constraints emerge from the interaction between specialized language networks and other cognitive systems (Fedorenko et al., 2024). On this processing-level interpretation, what speakers experience as difficulty may reflect demands distributed across specialized brain systems rather than a single cognitive bottleneck.

Dependency locality is the best-documented such cost. Integration difficulty rises with dependency length and peaks at loci of long-distance integration, with locality accounting for 59% of region-level variance in reading times in one experiment and 73% in another (Grodner & Gibson, 2005); the figure is considerably lower once item-specific variance is included. Because the effect is so robust, the formal treatment in §3 gives dependency locality a dedicated component rather than hiding it inside a residual term. High-locality constructions feel worse even when structurally well-formed, and because they're harder to produce and parse, speakers tend to avoid them.

That avoidance is a pathway from processing to change, developed formally in §2.5 and §4: lower-processing forms are easier to store and reuse, so simpler patterns can spread and conventionalize. The processing evidence supports this as a plausibility claim about one route into stability, not a direct neural explanation of grammatical change.

Processing constraints also interact with other components. A construction that's marginal due to low situational acceptance may become completely unacceptable when combined with processing difficulty. Conversely, highly entrenched constructions may remain acceptable despite considerable processing demands; strong conventionalization can partially offset processing costs.

Socio-pragmatic indexicality. The value of a construction extends beyond semantic content to include socio-pragmatic dimensions: how linguistic forms index aspects of social context, speaker identity, group membership, stance, or interpersonal relationships (Eckert, 2012; Silverstein, 1976).

In many varieties of Latin American Spanish, for example (e.g. the Río de la Plata region), speakers use *vos* and its associated verb forms instead of the *tú* forms used elsewhere in the Spanish-speaking world (Bertolotti, 2016):

(16) *¿Vos querés un café?*
you.SG want.2SG-VOS a coffee
'Do you want a coffee?'

Here, the use of *vos* rather than *tú* denotes the second-person singular hearer (the person being offered a coffee) and indexes the speaker's regional identity and familiarity with the local dialect. This indexical value may convey closeness, solidarity, or membership in a particular geographic and social community.

This situational view of grammaticality can manifest asymmetrically. Speakers from the Río de la Plata region may view a conversational situation as accommodating both their own norms and those of *tú*-using interlocutors; the situation itself can encompass both *vos* and *tú* as grammatical options. But speakers from *tú*-only regions might conceptualize the same situation more restrictively, defining it in a way that categorically excludes *vos* as a grammatical possibility. This asymmetry doesn't depend on different understandings of the forms themselves, but can arise from different ways of defining what the communicative situation allows, influenced by the indexical values attached to *vos* versus *tú* in their respective communities.

Phonology can carry constructional indexical value, but it bears on grammaticality only when an operator value conflicts. Babel (2025) provides an example. In a study conducted in Bolivia, participants were presented with audio stimuli where only the vowels were manipulated to reflect either a highland or lowland accent. This presented participants with incongruent identity cues (e.g. vowels from one accent along with consonants from the other). But this didn't trigger feelings of ungrammaticality. Instead, vowel contrasts activated expectations about consonant features and discourse markers, and some participants "hallucinated" identity-linked features that weren't present in the signal.

So a construction's value reaches beyond semantic features to socio-pragmatic aspects that shape how speakers and hearers negotiate authority, identity, solidarity, and other interpersonal relations. Recognizing these indexical dimensions helps explain why certain forms feel natural and grammatical to some speakers and out of place or even ungrammatical to others.

Situational entrenchment and acceptance. Even well-formed constructions with clear meanings may be judged ungrammatical if they lack conventionalized acceptance. This component captures the degree to which a form–value relation has become entrenched within relevant communicative situations.

(17) * *We sheared three sheeps.*

The regular plural marking is semantically transparent and structurally parallel to other English plurals, but the irregular form *sheep* is entrenched across English-speaking communicative situations, making the regularized version unacceptable. Without a metaphorical or playful justification (as in *the black sheeps of the family*, where the irregular plural might signal a figurative usage), the utterance is ungrammatical.

Situational entrenchment operates independently from the other components. A construction may have perfect value compatibility and minimal processing demands but still be rejected because a different form has been conventionalized for that meaning. Extreme rarity shows the point clearly. Some constructions are so infrequent that speakers lack a shared consensus about their status.

The independent relative genitive pronoun *whose* provides an example, being so unusual that Hankamer and Postal (1973) deem it non-existent. The contexts that license this construction require the simultaneous convergence of distinct pragmatic and syntactic conditions (sufficient accessibility of both possessor and possessum, the appropriate information structure, and an environment allowing ellipsis), which are seldom met all at once (Reynolds, 2024):

(18) ? *I saw Joan, a friend of whose was visiting.*
(adapted from Huddleston & Pullum 2002, p. 472)

In the COCA check reported here, no token of the construction appears (Davies, 2024). That absence supplies little negative evidence, because the opportunity set is tiny: contexts requiring an independent relative genitive are vanishingly rare. Speakers have little evidence either way: neither positive tokens nor the systematic absence that would accumulate if the construction were regularly passed over. The evidence therefore supports high *uncertainty* rather than confident rejection. Some speakers can draw an analogy with similar constructions and accept examples like (18); others remain unsure; still others, lacking sufficient exposure, can't construct a stable form–value relation at all. Even among those who grasp the construction analytically, Shain et al. (2020)'s account predicts that its unexpectedness would trigger high surprisal, compounding the uncertainty with processing difficulty.

This contrasts sharply with preempted cases like left-branch extraction, discussed next, where the opportunity set is large and the construction is systematically untaken, giving speakers overwhelming evidence that it isn't licensed.

Apparent categorical exclusions. Some constructions face rejection so persistent that it *appears* categorical. Left-branch extraction is the textbook case:

(19) * *Which did you buy* [__ *car*]*?*

The intended meaning is easily grasped; nothing in the semantics or pragmatics prevents understanding the speaker's intent. Nor does the construction seem excessively complex to process. Yet English speakers categorically reject it, and Snyder (2022, p. 22) treats this kind of pattern as a core case of "ungrammaticality".

Under OVMG, such patterns are analyzed not as cases that require a separate constraint on possible structure but as preempted gaps: situational licensing is driven toward zero by persistent preemption. Because the communicative job is common and an established competitor always wins (e.g., *Which car did you buy?*), speakers accumulate overwhelming evidence that the construction isn't part of the repertoire. Additional processing penalties, such as systematic garden-path reanalysis when the bare interrogative phrase is encountered, can strengthen the subjective categoricality without any independent structural constraint being posited. The formal dynamics appear in §3.

Cross-linguistic comparison doesn't threaten this analysis, because the dynamics are defined over a language's own niches. Left branch is a descriptive class tied to a particular phrase-structure analysis, not a comparative concept (Haspelmath, 2010); whether discontinuous nominals in case-rich, article-less languages instantiate the same configuration as the English case is far from clear, and the companion corpus study deliberately brackets the comparison (Reynolds, 2026d). What the model inherits instead is a prediction about opportunity structure: superficially similar discontinuous-nominal patterns should be learnable where inflectional marking makes the dependency recoverable, a productive source of discontinuous order supplies scaffolding, and a discourse niche makes the separated element useful. English meets none of the three, so the candidate has no scaffolding of its own and the contiguity-preserving competitors win every opportunity.

Several diagnostic criteria help identify these preempted-gap constructions:

1. Persistent unacceptability: The construction never gains acceptance; familiarity and repetition don't shift judgments.
2. Categorical rejection: Speakers find the constructions simply impossible, with no intermediate ratings.
3. Resistance to satiation: Unlike processing-heavy cases, repeated exposure doesn't improve acceptability.

Prediction: If participants are led to treat presented tokens (like LBE) as intentional dialectal or ingroup resources rather than errors, then satiation (or at least reduced feeling of ungrammaticality) should be more likely than in standard paradigms where the construction is treated as simply wrong.

If a construction resists even this preempted-gap analysis (a form for which no structural analysis is available at all), it would mark the residual case for an independent constraint on possible analysis. Treating classic cases like LBE as driven to zero under preemption keeps the constitutive core minimal and pushes the explanatory work into the dynamics of situational licensing.

WINNERLESS CELLS: PARADIGM DEFECTIVENESS. Preempted gaps have a winning competitor. Defective paradigm cells don't. They leave a high-opportunity cell empty without any single form winning it: most English varieties lack a negated contracted form for *am* (**amn't*, though Irish English and some Scottish varieties license it), and Russian speakers avoid the expected first-person singular non-past of *pobedit'* 'win', resorting to periphrasis instead (Broadbent, 2009; Pertsova, 2016; Pertsova & Kuznetsova, 2015; Sims, 2023; Thoms et al., 2025).

The niche is common and candidate forms are transparent, but no variant accumulates enough positive evidence to settle as the form. The source can be structural uncertainty, lexical restriction, learning dynamics, or avoidance, with the mixture varying across cases (Sims, 2023). In this account, defectiveness is a third licensing profile: high opportunity with evidence divided among candidate forms, leaving each candidate's posterior low in mean but unstable, rather than concentrated near zero.

The phenomenology matches: speakers report not knowing how to say it (ineffability), not the confident "that's wrong" of a preempted gap. The satiation-framing prediction also follows: defective cells should behave like low-evidence forms, movable under framing, not like preempted gaps.

2.2 DIAGNOSING (UN)GRAMMATICALITY AT A GLANCE

For exposition the recurrent instability modes can be read as a short decision tree (the formal definitions appear in §3):

Condition	⟶ Canonical outcome
no viable structural analysis	⟶ NONSENSE (**Can the have running*)
semantic/indexical clash	⟶ value incompatibility (**I've finished it yesterday*)
rarely encountered, low certainty	⟶ community-novel (*friend of whose*)
systematically preempted	⟶ preempted gap (left-branch extraction)
high opportunity, candidates split	⟶ defective cell (**amn't*)
high processing cost, structure recoverable	⟶ transient ill-formedness (centre embedding)
otherwise	⟶ grammatical

2.3 PATTERNS OF (UN)GRAMMATICALITY

When these components interact, constructions succeed or fail in systematic but non-binary ways. The diagnostic categories above name the main routes to failure; the decision tree summarizes them. Forms also differ in how far they fall from licensing.

DEGREES OF VIOLATION. The degree of violation reflects the scope and intensity of component mismatches. Violations can range from mild, easily recoverable deviations to complete breakdowns in interpretability.

Minor violations might involve a single component with partial conflict, as when a construction has moderate processing difficulty but full value compatibility and strong situational entrenchment. Such cases often receive intermediate acceptability ratings and may improve with exposure.

Severe violations typically involve multiple components or complete failure of a single critical component. In a construction lacking any viable form–value relation, the breakdown is total, while violations of categorical constraints produce consistent, strong rejection regardless of other factors.

The interaction between components can amplify or mitigate violations. Strong situational entrenchment can partially offset processing difficulty, while semantic transparency can make marginal syntactic patterns more acceptable. Conversely, multiple minor violations can compound to produce strong ungrammaticality judgments.

2.4 The subjective experience of grammaticality

The constitutive variables determine conventional grammatical status (§3.3); speakers register putative violations through feelings and judgments that can diverge from it. [2]

The feeling of ungrammaticality. The subjective feeling of ungrammaticality is a metacognitive response to linguistic violations. It parallels other well-studied metacognitive feelings like the feeling of knowing (FOK; Hart, 1965), which arises when one feels certain of knowing something but can't retrieve it.

There isn't a positive feeling of grammaticality, just as there isn't a feeling of having enough oxygen, only the negative feeling registered in its absence. The feeling of ungrammaticality, then, can be seen as the negative response triggered when an utterance violates expected form–value patterns.

This notion echoes Edward Sapir's "form-feeling", an often unconscious grasp of language patterns (Sapir, 1921, 1927). Evidence from aphasia supports the same separation. Patients with Broca's aphasia, who produce agrammatic speech, often exhibit self-monitoring behavior, attempting self-correction and expressing frustration with their grammatical errors (Oomen et al., 2005).

Unstable or missing form–value relations, detected during processing, can trigger this response. The response is a speaker-level heuristic for detection, not the definition of grammaticality. That separation explains why marginal constructions elicit gradient certainty, why repeated exposure can attenuate negative responses without necessarily changing status, and why objectively licensed constructions can still feel wrong under processing pressure. In that sense, (un)grammaticality can be illusory (Fanselow, 2021).

Distinguishing grammatical status from subjective ratings. Grammaticality research has to distinguish grammatical status, the conventional, population-level property made precise in §3.3, from the subjective feelings and ratings that speakers provide. (I use "objective" below only in that sense, population-level and convention-constituted, not judge-independent.)

Table 2 summarizes the two levels of analysis.

The measurement problem is familiar from other sciences: temperature can't be directly observed

[2] The distinction between grammatical status and the subjective feeling has a philosophical analogue in Cavell's (1969) distinction between knowledge and acknowledgment. For Cavell, acknowledgment isn't a separate capacity from knowing but an inflection of it, bearing knowledge toward the situation that produced it. Speakers don't merely *know* that a string is ungrammatical; they *register* it as a violation, with metacognitive consequences. For LLM-based grammaticality research, the distinction blocks an easy inference: LLMs can produce judgments that correlate with human ones, but whether they acknowledge the norms they're applying or merely process patterns that correlate with those norms remains an open question (Techio, 2026).

Table 2: Grammatical status versus subjective experience

Level	Nature	Observable through
Grammatical status	Whether a form–value relation is licensed in the communicative situation	Converging evidence: corpora, production, repair behaviour
Subjective feeling	Metacognitive warning signal that "something is wrong here"	Directly observable: ratings, eye-tracking, self-reports

but has to be inferred through the behaviour of thermometers. Grammaticality likewise has to be inferred from several measures, with acceptability ratings as one imperfect window. The same caution applies to language-model probabilities: raw string probability by itself doesn't separate grammatical from ungrammatical strings, and minimal-pair probability contrasts are informative only when the intended message is controlled (Hu et al., 2026).

The formal counterpart of this two-layer picture, including what it implies for interpreting the factor analyses that are common in experimental syntax, appears in the formal core (§3.7, §3.8): because such analyses target the structure of the subjective feeling, not grammatical status directly, a factor counts as evidence about the grammar only when it also shows up in the independent indicators for C_t, the population licensing state, and K, operator-value coherence.

This also explains why satiation and gradient judgments don't automatically threaten categorical status. A construction can remain unlicensed in the community's system while triggering progressively weaker negative responses through familiarization. Conversely, subjective response can remain continuous even when the population licensing state is effectively categorical (§4.4). Treating ratings as evidence rather than definition keeps grammatical hypotheses empirically vulnerable without collapsing the linguistic system of a communicative situation into individual responses to it.

Misattribution effects. Sometimes the subjective feeling of ungrammaticality doesn't accurately reflect objective grammatical status. Both false positives (grammatical constructions feeling ungrammatical) and false negatives (ungrammatical constructions escaping detection) occur systematically.

When grammatical constructions feel ungrammatical. Listeners and readers can misanalyze utterances, triggering feelings of ungrammaticality even though the construction conforms to standard patterns:

(20) *The old man the boats.* (Ritchie & Thompson, 1984)

On first reading, one might treat *old man* as a noun phrase and arrive at nonsense. But the sentence actually has *the old* as the subject (meaning 'the elderly'), and *man* as a verb. Interpreted correctly, the sentence means 'The elderly operate the boats', and is fully grammatical.

Processing overload provides another source of misattribution. Heavily nested structures like (15) are grammatically well-formed but trigger negative responses due to cognitive limitations rather than grammatical violations.

When ungrammatical constructions escape detection. Conversely, objectively ungrammatical utterances may fail to trigger negative responses when the intended meaning is strong:

(21) * *In Michigan and Minnesota, more people found Mr. Bush's ads negative than they did Mr. Kerry's.* (Pullum, 2009)

The intended interpretation is clear, and hearers readily infer the meaningful comparison. Under the relevant syntactic analysis, the comparison composes differently. The first clause suggests comparing numbers of people, but the second clause has *they* ('more people'), making the meaning 'more people did x than more people did y', which is nonsensical. But this violation often escapes detection.

Agreement errors can similarly hide within complex noun phrases:

(22) * *The patchwork of laws governing background checks, addressing assault-weapons limits, and regulating open-carry practices help explain why people continue to be wounded and killed.* (modified from Corbett, 2016)

The singular subject *patchwork* clashes with plural *help*, but the intervening plural nouns mask this violation. Cognitive resources devoted to processing complexity can prevent speakers from registering the agreement error.

2.5 Mechanisms of grammatical change

The stability of form–value relations isn't static. Various motivations alter the evidence balance for new and established forms, explaining how grammatical systems evolve over time. These aren't a loose catalogue of causes: each is a route by which the evidence balance of §4.3 can change sign, so each predicts a direction of change (which forms should gain licensing and which should lose it) and feeds the actuation dynamics of §4.5.

Semantic motivations for reanalysis. Speakers regularly deploy metaphors, analogies, and context-induced reinterpretations that stretch or shift the meaning of existing forms. Over time, such re-analyses can yield new grammatical constructions that weren't previously part of the language's repertoire.

A well-documented example is the development of the *going to* futurate construction. Historically, *going to* described physical motion toward a place:

(23) *I am going to London.*

Frequent usage in contexts where motion implied subsequent action (e.g. *I am going to fetch some firewood*) led to semantic shift. The directional component was reinterpreted as marking future intention rather than spatial goals:

(24) *It's going to rain.*

What once expressed physical trajectory now serves as a predictive futurate marker. The form–value relation changed through semantic reanalysis, driven by communicative utility.

Social motivations for accepting or rejecting forms. Linguistic variants often serve as markers of regional background, class, ethnic identity, or age group. These social cues motivate speakers to favor some forms over others; over time, those preferences shift what counts as grammatical.

The decline of the simple past (*le passé simple*) in modern spoken French illustrates social motivation. Forms like *il alla* ('he went') became associated with formal, literary, or provincial speech. To avoid signaling undesirable social affiliations, speakers gravitated toward compound forms like *il est allé*, which lacked these status-laden connotations.

Situational licensing tracks changing social dynamics: forms that were once fully grammatical can become socially marked and gradually excluded from normative grammar.

Structural motivations. Structural considerations arise from cognitive and communicative demands, including processing limitations, the need for clarity, and the influence of analogy.

Processing constraints and memorability. Transmission tends to favor patterns that minimize processing demands. Forms requiring multiple long-distance dependencies or heavy embedding are less likely to achieve stable transmission across generations. This pressure toward processability shapes grammatical evolution, with simpler patterns spreading through communities.

Analogical extension. When speakers encounter new communicative challenges, they often extend known patterns to new contexts:

(25) ? *I asked about what did she do.*

Though currently marginal, such forms appear to gain traction by analogy with main-clause interrogatives (*What did she do?*). If accepted, this analogical extension would represent structural motivation reshaping the grammar.

Iconic motivations. Forms are sometimes accepted precisely because their structure reflects their meaning. Reduplication for intensity provides a simple example:

(26) *a big big problem*

The formal repetition iconically mirrors the magnitude being expressed. Such patterns are readily interpretable and likely to spread because the form–value link is immediately apparent.

Iconicity remains syntactically constrained, though: it occurs in pre-head modifiers but is ungrammatical as predicative complement:

(27) * *the problem was big big*

These motivations (semantic, social, structural, and iconic) can interact and sometimes conflict. While Optimality Theory has productively modeled such competing pressures through ranked constraints (Prince & Smolensky, 2004), OVMG views their resolution as arising from usage within communicative situations rather than solely from fixed rankings. The specific weighting of motivations reflects historically established patterns and communicative needs.

3 A formal core: grammaticality as conditioned stability

The framework separates two explanatory targets: (i) a state theory specifying the conditions under which an utterance type is grammatical *for a population in a communicative situation* at time t; (ii) a dynamics module specifying how such states tend to arise and persist. The state theory specifies what makes a form grammatical at a time; the dynamics module explains how such grammatical states arise, persist, and change. In projectibility terms, the state theory specifies the profile, while the dynamics module states its support: what stabilizes the profile and, where the evidence warrants, what maintains it. Support claims come in grades that have to be earned separately. It is one thing to show that a profile stabilizes, another to show that a production–repair loop maintains it, and a still stronger thing to show that some process detects deviations and pushes the profile back. None of those stronger claims is inherited from the framework's name (Reynolds, 2026a, 2026b).

In Woodward's interventionist terms, a stabilizer earns that status when an intervention on it would change the profile in an invariant, predictable way (Woodward, 2001, 2003). Table 3 summarizes the notation.

Table 3: Notation summary

Symbol	Interpretation
State theory	
u	Utterance type or candidate token
$g \in \mathcal{H}$	Construction node in a hierarchy
$c \in \mathcal{C}$	Conditioning state (situation + norm-centre)
$\mathsf{map}(u, c)$	Structural viability ($\in \{0, 1\}$)
$K(u, c)$	Operator-value coherence with reject option ($\in [0, 1]$)
$C_t(u, c)$	Situational licensing at time t ($\in [0, 1]$)
$\widetilde{G}_t(u, c)$	Graded stability score ($= \mathsf{map} \cdot K \cdot C_t$)
Feeling model	
$F_{i,t}(u, c)$	Subjective feeling of anomaly
$L(u)$	Dependency locality cost
$\mathrm{R}_i(u, c)$	Processing cost vector (interference, locality, surprisal)
Dynamics (§4)	
$N_t(n, c)$	Number of opportunities for niche n in state c
$\rho_t^\star(v \mid n, c)$	Counterfactual choice probability among analogical candidates
s_t, e_t, p_t	Positive evidence, error evidence, preemption mass

3.1 Conditioning structure

Let $c \in \mathcal{C}$ be a CONDITIONING STATE, a construed communicative situation in Wiese's (2023) sense, together with whatever norm-centre is treated as relevant. Nothing here requires c to be externally given; interlocutors can misalign about c, and c can be learned, split, and renegotiated over time.

For many purposes it helps to think of c as a bundle that may include situational features (activity type, medium, footing), stable baselines tied to speaker ascription, and norm-orientation (discourse-community identification), but the formalism treats c abstractly as the conditioning variable that selects the relevant distribution of form–value relations.

3.2 Objects

Let u range over utterance tokens or fine-grained utterance types, with structural representation $M(u)$. Let $\sigma(u, c)$ be the utterance-level interpretation made available in c (lexical content plus pragmatic enrichment, indexical resolution, discourse update, etc.).

Utterance types inherit licensing from a hierarchy of constructional nodes $\mathcal{H}$. A novel token such as *She texted him the address* need not have been encountered before; it can still be covered by licensed nodes for ditransitive transfer, pronominal object order, and the relevant lexical-class relations. Let $A(u) \subset \mathcal{H}$ be the nodes activated by u, and let $\lambda_g(u, c)$ be non-negative coverage weights with $\sum_{g \in A(u)} \lambda_g(u, c) = 1$. Each node g has its own licensing posterior. A token-level licensing estimate can then be derived by partial pooling:

$$C_t(u,c) = \sum_{g \in A(u)} \lambda_g(u,c)\, C_t(g,c).$$

The sum is schematic; a similarity kernel over representations would do the same work. What matters is that the learner updates constructional regions, not only memorized sentence types. This prevents the dynamics from predicting that every first-heard sentence has zero licensing.

The framework distinguishes three state quantities.

(I) Mapping viability. Let $\mathsf{map}(u,c) \in \{0,1\}$ indicate whether structural analysis can yield at least one well-typed representation for u (category assignment + structural composition) throughout the derivation.

$$\mathsf{map}(u,c) = \begin{cases} 1 & \text{if } \exists \text{ valid analysis } M(u) \text{ yielding } \mu(M) \\ 0 & \text{otherwise.} \end{cases}$$

map is prior to *situational licensing* in one specific sense: it registers whether any covering analysis exists at all (word salad and category errors defeat it) while enforcing nothing about whether the covered analysis is licensed in c. Situational non-licensing belongs only in C_t. But map isn't pre-conventional; §3.4 spells out its dependence on the constructional hierarchy, stored wholes included.

(II) Coherence as satisfied concentration of interpretation. Even when $\mathsf{map}(u,c) = 1$, interpretation can be unstable: multiple incompatible construals may compete, or the best available construal may violate an operator constraint. A normalized distribution over ordinary interpretations would measure ambiguity but not violation: a single catastrophic construal would still receive probability 1. To avoid that bug, the model adds a null outcome ω_0 representing rejection or failed operator-value satisfaction. Let $\Omega^+(u,c) = \Omega(u,c) \cup \{\omega_0\}$ and fix $E(\omega_0; u,c) = \kappa_c$:

$$p(\omega \mid u,c) = \frac{e^{-E(\omega;u,c)}}{e^{-\kappa_c} + \sum_{\omega' \in \Omega(u,c)} e^{-E(\omega';u,c)}}.$$

$$K(u,c) = \max_{\omega \in \Omega(u,c)} p(\omega \mid u,c) \in [0,1].$$

I choose the null-outcome repair because it preserves the probabilistic interpretation of K. An unnormalized score $e^{-E(\omega^\star)}$ or an explicit $K_{\text{peak}} \times K_{\text{satisfaction}}$ decomposition would make similar predictions, but the null outcome keeps ambiguity and satisfaction in a single race. A good single construal has $E(\omega^\star) \ll \kappa_c$ and $K \approx 1$; a transparent but operator-violating construal has $E(\omega^\star) \gg \kappa_c$, so probability mass shifts to ω_0 and $K \approx 0$.

The energy model $E(\omega; u, c)$ is *open-ended*:

$$E(\omega; u,c) = \sum_{j \in \mathcal{J}(c)} w_j\, \phi_j(\omega; u,c),$$

where $\mathcal{J}(c)$ includes constraints on grammaticalized features live in c. In the demonstrations below, I assume a minimal inventory: (1) temporal alignment where tense–aspect is grammaticalized, (2) argument-structure satisfaction where a construction licenses roles, (3) information-structural fit where topic/focus packaging is grammatically specified, and (4) indexical consistency only when the contrast belongs to a closed accountable repertoire. This criterion makes K partly derivative of licensing facts. A value can enter E only after the community has conventionalized it as a grammaticalized constraint.

This decomposition bridges the narrative diagnostic categories of §2.1 to the formal variables. For instance, one can group coherence demands by subsystem, denoting the sum over purely semantic demands as E_{sem} and over indexical/pragmatic demands as E_{index}, to yield an expositional factorization $K \approx K_{\text{sem}} \cdot K_{\text{index}}$. In practice, the framework assumes that stability depends on the joint concentration of all constraints live in the communicative situation c.

(iii) Situational licensing (entrenchment) under conditioning. Each individual i maintains a graded licensing state for u in c: a credence $\pi_{i,t}(u, c) \in [0, 1]$ that u is a community resource there. §4.3 models this state as a Beta posterior. The licensing *attitude* $\Lambda_{i,t}(u, c) \in \{0, 1\}$ (treating the relationship as a legitimate option rather than a performance slip or nonce error) is a thresholded read-out of that graded state, not a separate primitive.

The population-level licensing state is the distribution of $\pi_{i,t}(u, c)$ across the population induced by c. Its mean,

$$C_t(u, c) = \mathbb{E}_i\big[\pi_{i,t}(u, c)\big] \in [0, 1],$$

is the object previously called "entrenchment", explicitly conditioned on c. The population quantified over isn't a separate primitive; it's induced by the norm-centre and communicative situation bundled in c.

The mean doesn't exhaust the state. A community in which half the speakers confidently license u and half confidently reject it, and a community in which every speaker sits at credence one-half, share $C_t = 0.5$ but differ in dispersion, both across speakers and within each speaker's posterior. Stable mixture and shared uncertainty are different conditions with different signatures (bimodal versus unimodal individual judgment distributions, high versus low within-speaker consistency), and §3.3.2 shows that some of the model's projections run over exactly that difference. Where only the mean matters, I write C_t; where concentration matters, I refer to the licensing state.

3.3 Grammatical status as a conditioned state property

I write $\mathbb{I}[\cdot]$ for the indicator function: $\mathbb{I}[\phi] = 1$ if ϕ is true and 0 otherwise.

The state theory distinguishes a graded stability score from a categorical membership predicate.

3.3.1 A graded stability score

Recall that $\mathsf{map}(u, c) \in \{0, 1\}$ indicates whether u has at least one available structural analysis in conditioning state c that yields a well-defined operator value. Let $K(u, c) \in [0, 1]$ be coherence as satisfied concentration of the interpretation distribution, and let $C_t(u, c) \in [0, 1]$ be the population licensing rate at time t in c.

Define the graded stability score:

$$\widetilde{G}_t(u, c) = \mathsf{map}(u, c) \cdot C_t(u, c) \cdot K(u, c) \in [0, 1].$$

This quantity is the workhorse for modeling gradience and for linking the state theory to behavioural measures.

3.3.2 Projectible use of the graded score

$\widetilde{G}_t$ isn't bookkeeping after the real explanation has happened. It's a macro-state over which grammaticality projects.

Some generalizations run over $\widetilde{G}_t$ as such. Production and repair behaviour at time $t+1$ depend partly on $\widetilde{G}_t$:

$$\Pr(\text{produce } u \mid n, c, t) \propto \widetilde{G}_t(u, c)\, \rho_t^{\star}(u \mid n, c),$$
$$\Pr(\text{repair } u \mid c, t) = r_0 + r_1\big(1 - \widetilde{G}_t(u, c)\big), \qquad r_1 > 0.$$

For these, a low score licenses the same expectations regardless of whether it comes from mapping failure, operator-value incoherence, or non-licensing, and high-$\widetilde{G}$ forms show the converse profile.

Other generalizations need the finer state: the licensing posterior with its concentration, not just its mean. Two constructions can share a low C_t while differing in how much evidence backs it, and the update rule (28) makes them behave differently under new exposure: a diffuse posterior should move with framing and familiarization, while a concentrated one shouldn't. Satiation profiles, framing sensitivity in preempted-gap cases, and actuation (§4) are projections over the posterior state. This separates the independent relative *whose* (low evidence, diffuse posterior) from left-branch extraction (heavy preemption, posterior concentrated near zero): the two can approach similar means while projecting differently.

The feedback loop closes as before. Production and repair generate the later evidence streams s_t, e_t, and p_t, so individual behaviour updates population licensing, and the resulting population state projects back into future production and repair. The projectible generalizations are satiation profiles, transmission fidelity, repair rates, and change trajectories: some over $\widetilde{G}_t$ as such, some over the posterior state that $\widetilde{G}_t$ summarizes.

Figure 2 makes the two-dimensional state visible. The horizontal axis is the licensing mean, which $\widetilde{G}_t$ reads; the vertical axis is the posterior's concentration, which it discards. Preempted gaps and defective cells share a low mean and separate only on the vertical axis, so any account that projects from the mean alone can't tell confident rejection from unresolved uncertainty.

3.3.3 A categorical membership predicate

Communities often treat grammaticality as a constitutive membership fact: either a relationship counts as an available resource in c or it doesn't. A categorical predicate can be defined by thresholding the graded score:

$$G_t(u, c) = \mathbb{I}\Big[\widetilde{G}_t(u, c) \geq \tau(c)\Big]\,.$$

The threshold $\tau(c)$ isn't a hidden grammatical parameter. It's a decision criterion associated with the communicative situation: some situations are strict about what counts as "in the repertoire", others are permissive.

3.3.4 A decision-theoretic motivation

Suppose a judge in c is choosing between two labels, $\ell \in \{\textsc{in-repertoire}, \textsc{not-in-repertoire}\}$, and faces asymmetric losses. Let $L_{\text{FA}}(c)$ be the loss of treating an item as IN-REPERTOIRE when it isn't, and $L_{\text{FR}}(c)$ the loss of treating an item as NOT-IN-REPERTOIRE when it belongs there. Under

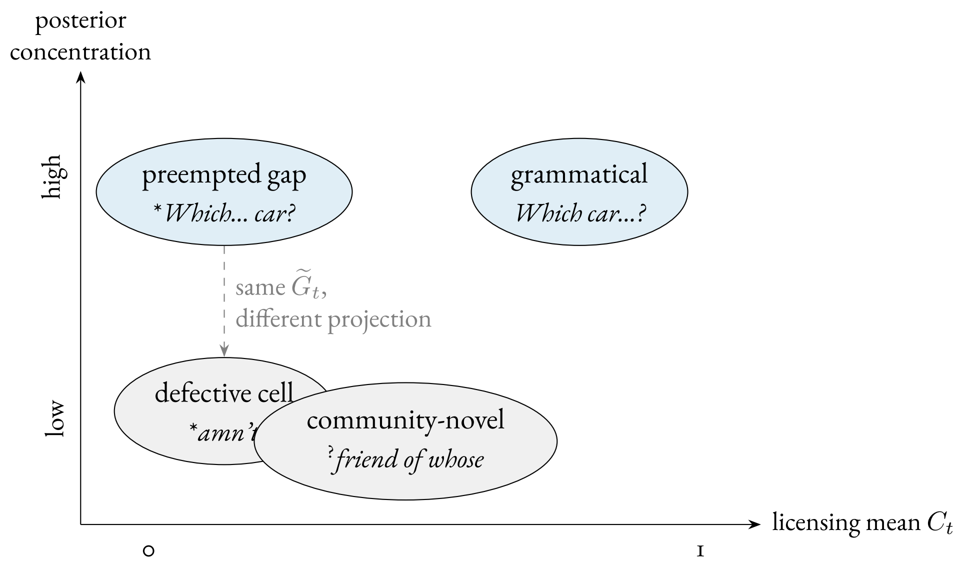

Figure 2: The licensing state has two dimensions, but the mean-product $\widetilde{G}_t$ reads only the horizontal one. Preempted gaps (dense preemption, posterior concentrated near zero) and defective cells (split candidates, posterior low but diffuse) share a licensing mean and part only on concentration, the axis that predicts satiation, framing sensitivity, and actuation.

the simplifying assumption that $\widetilde{G}_t(u, c)$ is a calibrated signal of stability, the optimal rule is to accept u as IN-REPERTOIRE whenever:

$$\widetilde{G}_t(u, c) \;\geq\; \tau(c) \;=\; \frac{L_{\text{FA}}(c)}{L_{\text{FA}}(c) + L_{\text{FR}}(c)}.$$

On this view, $\tau(c)$ is induced by payoff structure: high-stakes institutional contexts raise L_{FA} (and thus τ), whereas in-group, low-stakes contexts raise L_{FR} (lowering τ).

Two clarifications keep the categorical predicate from overclaiming. First, calling the status "objective" means population-level and convention-constituted, not judge-independent in any stronger sense. The status stands apart from any individual's response the way facts about word meaning or currency do; it's worldly in that, given the community's practices, the projected relations aren't made available by the analyst's decision to group cases together (Reynolds, 2026a). CONVENTIONAL STATUS would be an equally accurate name, and I use the two interchangeably below. Second, the threshold's arbitrariness is confined by the dynamics: where licensing posteriors are bimodal (§4.4), G_t is insensitive to the exact value of $\tau(c)$ and the categorical predicate does its work; in the contested middle, judgments should be unstable and stakes-sensitive, which is a prediction rather than an embarrassment. The categorical claim is thus a scoped one: it holds where the population state is bimodal and is silent elsewhere.

3.4 What map does and doesn't do

The mapping predicate $\mathsf{map}(u, c)$ is reserved for genuine analyzability failure: cases where no structural analysis is available that yields any well-defined operator value in c. This gives the only categorical failure mode in the representational sense.

Analyzability is relative to the constructional hierarchy $\mathcal{H}$, stored wholes included, not to productive composition alone. Syntactically anomalous idioms make the difference visible: *by and large*, *far be it from me*, and *come Monday* have no analysis by the productive syntax of English, but they're fully conventional (Fillmore et al., 1988). If map meant "parsable by general rules", these would come out $\mathsf{map} = 0$ and hence ungrammatical, the wrong result. They're covered instead by their own stored nodes in $\mathcal{H}$: coverage by some licensed node, productive or memorized, is what analyzability means here. The cost of this choice is candour about status: map isn't pre-conventional. What it contributes is a different *kind* of failure (no covering node at all), not a convention-free one.

Cases that motivate talk of categorical structural exclusion aren't treated here as mapping failures. In those cases, the intended analysis is typically available as a candidate representation, making $\mathsf{map}(u, c) = 1$, and the intended construal can be coherent once stipulated (so $K(u, c)$ need not be small). What makes the pattern behave categorically in the community is instead the licensing term: under the relevant conditioning states, the community converges on near-universal non-licensing, $C_t(u, c) \approx 0$. This gives the formal counterpart of what earlier drafts call "near-zero entrenchment".

3.4.1 Assigning a failure to map, K, or C_t

Because K's coherence constraints, the set $\mathcal{J}(c)$, are themselves derivative of licensing facts (§3.2), the three quantities aren't independent primitives but ordered, and the multiplicative form $\mathsf{map} \cdot K \cdot C_t$ has to be read with that order in view. The operator inventory conventionalized in c fixes which features are live in $\mathcal{J}(c)$; only then is a candidate's coherence score K well posed; and C_t, the population licensing state, records whether the population licenses the specific relation. That ordering yields pre-judgment diagnostics for assigning a failure, applied before consulting acceptability data. Ask first whether any node in $\mathcal{H}$ covers u: if none does, the failure is map (as in **Can the have running*). If a covering node exists, ask whether the best available construal violates a feature already live in $\mathcal{J}(c)$: if so, the failure is K, as in **I've finished it yesterday*, where the perfect's temporal feature is live in English and the past adverb violates it. If a covering node exists and the construal violates no live feature, but the population withholds the relation for that communicative job, the failure is C_t, as in **I have 25 years* for age, where the *have*+years frame is coherent but unlicensed for that job. Each branch is checkable against the independent indicators of §3.6 rather than read off the judgment it explains, which is what keeps the three-way assignment from reproducing the circularity a pure community-verdict account would incur.

3.5 Interlocutor misalignment about conditioning

Because c is construed, interlocutors can disagree about which conditioning state is in force. Let a hearer h maintain a posterior $q_h(c \mid e)$ over conditioning states given cues e (situational cues, ascription cues, stance cues, etc.). Then the hearer's expected stability score is

$$\widetilde{G}_t^{(h)}(u \mid e) = \sum_{c \in \mathcal{C}} \widetilde{G}_t(u, c)\, q_h(c \mid e).$$

3.6 Identifying the conditioning state before prediction

The conditioning state c isn't a free parameter to be selected after a judgment is observed. For empirical work, c has to be fixed before predicting ratings or repair behaviour. The posterior $q_h(c \mid e)$ is the identification device: it's estimated from non-judgment cues such as activity type, medium, participant roles, speaker ascription, register cues, and overt norm orientation. A form like *might* *could* isn't rescued by choosing a sympathetic norm-centre after the fact; the model predicts different judgments only when the independent cues make that norm-centre probable.

This also blocks a competence/performance redux. C_t, the population licensing state, is identified by corpus-rate-per-opportunity, elicited production, and repair behaviour, not by acceptability ratings alone. F, the subjective-feeling variable, is identified by ratings, online measures, and self-reports. The model is falsified where these dissociate in the wrong direction: if satiation changes later corpus rates for a purportedly preempted gap, if repair behaviour tracks prescriptive dissonance rather than licensing, or if high-opportunity preempted forms behave like low-evidence uncertainties, the state assignment is wrong.

Reassignment isn't free, though. Localizing a failed prediction to a narrower conditioning state is warranted only when that state is specifiable independently of the surviving cases, by activity type, medium, participant roles, or norm-orientation fixed before the failures were seen (Reynolds, 2026a). A partition of $\mathcal{C}$ redrawn around exactly the successes redescribes failure as success; it doesn't rescue the model. Confirmatory studies should register the conditioning partition and the estimation protocol for $\rho_t^\star$ (e.g. forced-choice norming over the competitor set, estimated before the licensing data is touched). The corpus checks in this paper are illustrative probes, not registered tests. Under that discipline, repeated post-hoc reassignment across future studies counts against the framework itself, not just against particular state assignments.

Two identification points deserve to be explicit, because a corpus rate estimates the product $C_t \cdot \rho_t^\star$ (§4.2), not C_t alone. First, $\rho_t^\star$ has to be estimated from evidence that doesn't include the licensing data it later interprets. A forced-choice norming task supplies it: present the niche n and the competitor set $\mathcal{V}_{n,t}^\star$ to a separate sample and record how often each variant is chosen when stipulated to be available. That yields $\rho_t^\star$ before any corpus rate for u is consulted, so a low rate can be diagnosed as non-licensing ($C_t \approx 0$, $\rho_t^\star$ high) rather than licensed-but-dispreferred selection (C_t high, $\rho_t^\star \approx 0$). Second, $\tau(c)$ isn't fit to the judgments it predicts. Where the licensing posterior is bimodal (§4.4), G_t is insensitive to the exact threshold, so no point estimate of $\tau(c)$ is needed; in the contested middle, $\tau(c)$ is set by independent stakes cues (institutional versus in-group footing), and the movable-cutoff prediction of §3.3 tests it directly: manipulate the stakes and the location of any discontinuity should move, which can't happen if $\tau(c)$ were merely a curve fitted to the responses.

3.7 The feeling of (un)grammaticality

Let $F_{i,t}(u,c) \in (-1,0]$ be the subjective anomaly signal for individual i. Model it as a bounded response to an unbounded evidence total. The raw signal accumulates low stability, implementation costs, and ideological overlays:

$$A_{i,t}(u,c) = \alpha\big(1 - \widetilde{G}_t^{(h)}(u \mid e)\big) + \gamma^\top \mathrm{R}_i(u,c) + \delta\, D_i(u,c) + \varepsilon_i,$$

$$A_{i,t}^+(u,c) = \max\{A_{i,t}(u,c), 0\},$$

$$F_{i,t}(u,c) = -\frac{A_{i,t}^+(u,c)}{1 + A_{i,t}^+(u,c)},$$

where $h = i$ is the hearer processing the signal, and $\widetilde{G}_t^{(h)}$ is the hearer's expected stability score accounting for situational uncertainty (§3.5). The bounded link keeps F in $(-1,0]$ however large the summed costs grow, and the floor at zero encodes the absence of any positive feeling of grammaticality

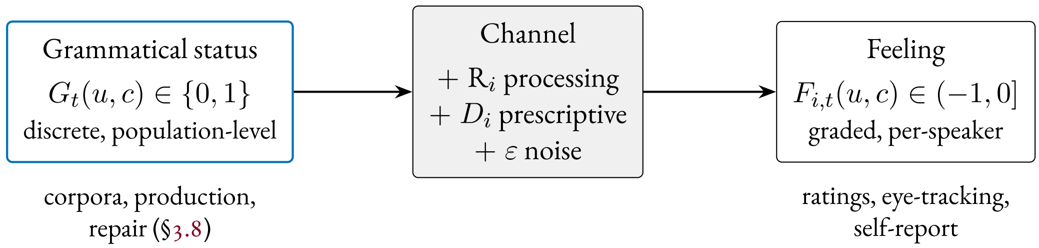

Figure 3: The two-layer architecture. Objective, discrete grammatical status isn't observed directly; it's filtered through processing costs, prescriptive dissonance, and noise into the gradient feeling that ratings measure. The two layers are identified by different evidence streams, which is what lets them dissociate and so falsify each other (§3.6).

introduced above. $\mathrm{R}_i(u,c)$ is a vector of processing/repair costs (locality, interference, garden-path reanalysis, surprisal, etc.). The locality component $L(u)$ sums over the dependencies $d \in \mathcal{D}(u)$:

$$L(u) = \sum_{d \in \mathcal{D}(u)} \ell\big(|d|\big), \qquad \ell(k) = \begin{cases} k, & k \le K_{\text{sat}}, \\ K_{\text{sat}} + \beta\,(k - K_{\text{sat}})^{\eta}, & k > K_{\text{sat}}. \end{cases}$$

Here $|d|$ is the length of dependency d (in intervening discourse referents or words), K_{sat} is a saturation threshold beyond which additional distance contributes sublinearly, and β, η (with $0 < \eta < 1$) control the rate of that sublinear growth. $D_i(u,c)$ is prescriptive dissonance.

Interpretation. $F = 0$ corresponds to a null affect (no anomaly reported), while $F < 0$ is a metacognitive warning proportional to the summed evidence of misfiring. The same feeling can arise from different mixtures of causes (structural, procedural, or social).

Measurement strategy. F can be measured via explicit ratings (mapped monotonically to $(-1,0]$) or implicit probes like N400/P600 magnitudes and self-paced reading slowdowns (proxies for components of R_i).

Figure 3 collects the two layers. The discrete, population-level status G_t is filtered through the processing costs R_i, prescriptive dissonance D_i, and noise into the gradient, per-speaker feeling F that ratings record. Because the two layers are read from different evidence streams, they can dissociate, which is what makes the account falsifiable (§3.6).

3.8 Measurement model for situational licensing (C_t)

$C_t(u,c)$, the population licensing state for u in c, is latent and is estimated from converging indicators rather than from ratings. Let $\mathrm{y}_t(u,c)$ be a vector of observable traces (corpus rate per opportunity, elicited production probability, repair probability, and recognition as a legitimate resource). Repair probability enters with the sign predicted by §3.3.2: resources with low C_t (weak population licensing) should invite repair when participants orient to the relevant norm-centre. A minimal measurement model is

$$\mathrm{y}_t(u,c) = \mathrm{b} + \boldsymbol{\Lambda}\,\mathrm{logit}\big(C_t(u,c)\big) + \boldsymbol{\epsilon}.$$

The logit is well defined because C_t never reaches the boundary: finite Beta updating (§4.3) keeps the posterior mean strictly interior, so the values written $C_t \approx 0$ or $C_t = 1$ in the prose and the worked examples denote near-boundary posteriors, not literal endpoints.

What factor analysis reveals. When researchers conduct factor analyses on acceptability ratings, a common approach in experimental syntax, they necessarily target the structure of the subjective feeling, not grammatical status directly. The factors that emerge reflect both genuine grammatical constraints and additional sources of variance that modulate subjective responses without affecting grammatical status, so each maps onto the model's variables only under stated conditions:

- Operator licensing: When this factor loads heavily on rejected items and converges with corpus-rate-per-opportunity, elicited production, and repair behaviour, it reflects C_t, population licensing.
- Value coherence: This factor, including information-structure clashes, belongs to K, operator-value coherence, only when it tracks constraints on grammaticalized operator values.
- Situational entrenchment/frequency: High-frequency patterns that speakers have conventionalized provide evidence about C_t, population licensing, but raw frequency has to be normalized by opportunity set.
- Sociolinguistic appropriateness: This factor has mixed status. Indexical clashes can affect C_t when they enter a closed accountable repertoire, but stylistic infelicities typically affect F.
- Processing cost/memory load: This factor belongs to F unless it feeds back into production, repair, and later licensing through the loop in §3.3.2.
- Morphophonological well-formedness: When phonological patterns realize obligatory operator exponents, violations affect K or C_t. Otherwise, they influence the subjective response.

The methodological upshot is a filter on evidence. Research that aims to uncover the grammar should model a factor only when it also appears in the independent indicators for C_t (population licensing) and K (operator-value coherence); research into judgment behaviour should retain the full factor structure, since the subjective feeling is precisely what predicts how speakers rate sentences presented out of context.

The corpus-rate-per-opportunity indicator is concrete. Figure 4 shows one worked case: agreement on collective-partitive subjects (*a bunch of people*, *the majority*), where surface-head number (singular) and notional number (plural) diverge. Counting only subject-position tokens after KWIC filtering (so non-subject false positives are removed) gives a plural-agreement share with a Wilson interval for each cell. The shares sit far above parity in every cell, with the interval width tracking the token count rather than the direction. This gives a licensing estimate for a contested operator contrast read off usage per opportunity instead of ratings: a latent C_t recovered from a converging indicator, with its uncertainty made explicit.

Table 4 makes the figure auditable: exact filtered cell counts, so the estimate can be checked rather than taken on trust.

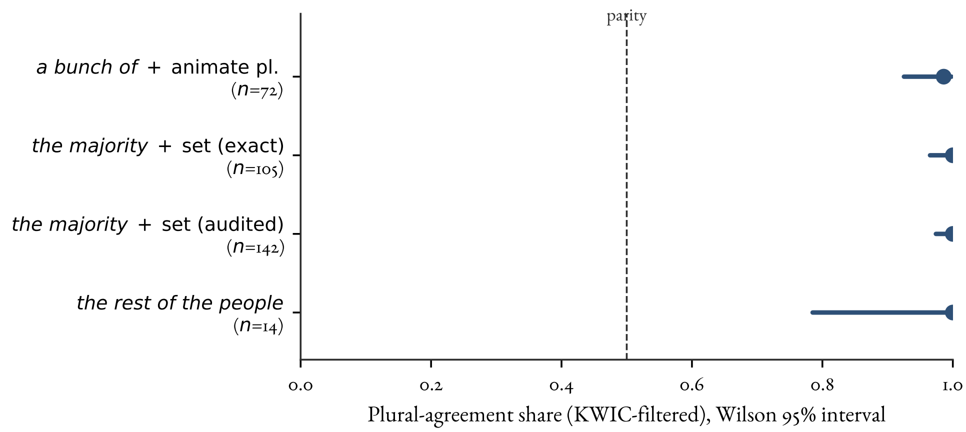

Figure 4: Corpus-rate-per-opportunity as an estimator of C_t for a contested agreement contrast. Plural-agreement share (dot) with Wilson 95% interval (bar) for collective-partitive subjects in COCA, after KWIC filtering to subject-position tokens; the dashed line marks parity (no directional preference). Data: `agr-coca-projection` probe (*bunch*, *majority/minority*, and partitive cells). Notional plural dominates in every cell, and the interval widens with the smaller denominators rather than drifting toward parity.

Table 4: Filtered agreement cells behind Figure 4. Counts are KWIC-filtered subject-position tokens in COCA; "pl." and "sg." are plural- and singular-agreement targets. The surface-head baseline predicts singular; the observed share is plural.

Cell	pl.	sg.	N	Plural share (Wilson 95%)
a bunch of + animate pl.	71	1	72	0.986 (0.925–0.998)
the majority + set (exact)	105	0	105	1.000 (0.965–1.000)
the majority + set (audited)	142	0	142	1.000 (0.974–1.000)
the rest of the people	14	0	14	1.000 (0.785–1.000)

3.9 Worked examples: deriving status from components

The status of the four canonical examples (§2.2) can now be defined by computing their component values.

1. **Can the have running* (Nonsense): `map` $= 0$ because structural analysis fails to assign a valid category structure. Result: $\widetilde{G} = 0$. Status: Ungrammatical.
2. **I've finished it yesterday* (Semantic Clash): `map` $= 1$ (well-formed syntax), $C_t(g, c) \approx 1$ at the level of the component constructions (perfect, past adverbial), but $K \approx 0$ due to the clash between Present Perfect and specific past adverbial (temporal alignment ϕ_{temp} fails). Result: $\widetilde{G} \approx 0$. Status: Ungrammatical.
3. *?friend of whose* (Situationally-Novel): `map` $= 1$, $K \approx 1$ (perfectly interpretable), but the licensing posterior stays diffuse: the opportunity set N_t is tiny, so preemption mass p_t barely accumulates and the posterior remains near its prior with wide dispersion, meaning "not established"

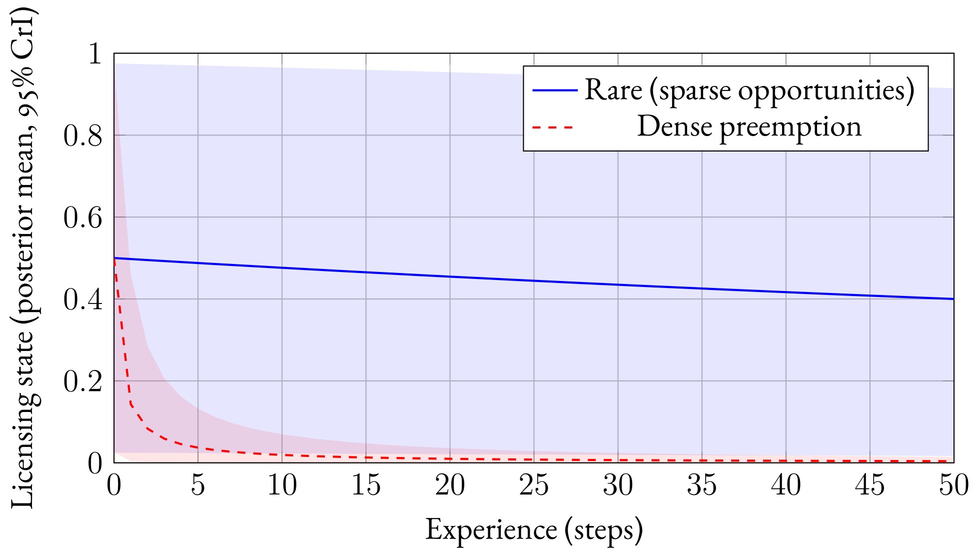

Figure 5: Licensing posteriors under preemption, from a Beta$(1, 1)$ prior with per-step effective preemption mass 0.01 (rare, blue) versus 5 (dense preemption, red); lines are posterior means, shaded regions 95% credible intervals. The contrast is one of kind, not just rate: the rare construction stays near its prior with a credible interval spanning nearly the whole unit interval (indeterminacy), while the dense-preemption posterior both falls and concentrates, collapsing onto confident rejection. The projections in §3.3.2 that turn on evidence structure (satiation, framing sensitivity, actuation) run over this dispersion difference, which the means alone don't encode.

rather than rejected. Result: $\widetilde{G}$ intermediate, low-confidence. Status: Marginal/Undefined.

4. **Which did you buy car?* (Preempted Gap): $\texttt{map} = 1$, $K \approx 1$, but the licensing posterior is driven toward zero and concentrates there. Since N_t is large (questions are frequent) and the counterfactual choice probability $\rho_t^\star$ is high, the effective preemption mass p_t is large, collapsing both the posterior mean and its dispersion (Figure 5). Result: $\widetilde{G} \approx 0$, high-confidence. Status: Categorically Ungrammatical.

4 Dynamics: why the state stabilizes

The dynamics module explains trajectories of $C_t(u, c)$ (and, in principle, refinements of the conditioning partition itself). It doesn't define grammaticality.

4.1 Niches, competitors, and opportunity sets

Let n index a CONSTRUCTIONAL NICHE (a communicative job), and let $\mathcal{V}^\star_{n,t}$ be the opportunity set: the observed, licensed, and analogically generated variants that could plausibly do that job in some conditioning states. Let $N_t(n, c)$ be the number of opportunities for niche n in conditioning state c over some time window, and $k_t(v, n, c)$ the observed count of variant $v \in \mathcal{V}^\star_{n,t}$.

The star matters. Preempted gaps and innovations often involve variants with no positive tokens. Their counterfactual utilities are still defined because the hierarchy $\mathcal{H}$ generates analogical candidates. English learners can represent a left-branch extraction candidate by combining interrogative fronting, modifier–noun packaging, and question formation, even if no one licenses that candidate as an English resource. The same machinery lets a novel but ordinary sentence inherit coverage from licensed higher-grain constructions rather than being preempted merely because that exact sentence type is new.

4.2 Usage as licensing × choice among candidates

Separate LICENSING from SELECTION. A variant can be licensed but rarely chosen.

Let $\rho_t^\star(v \mid n, c) \in [0, 1]$ be the counterfactual probability of choosing v if it were available in niche n under c. A flexible choice model is a softmax over utilities:

$$\rho_t^\star(v \mid n, c) = \frac{e^{U_t(v;n,c)}}{\sum_{v' \in \mathcal{V}_{n,t}^\star} e^{U_t(v';n,c)}}, \qquad U_t(v; n, c) = \boldsymbol{\theta}^\top \mathrm{f}(v; n, c).$$

At the population level, the expected usage rate $\pi_t(v \mid n, c) \approx C_t(v, c) \cdot \rho_t^\star(v \mid n, c)$. The counterfactual choice term can be high even when $C_t(v, c)$ is near zero; that is exactly the condition under which absence becomes informative.

4.3 Dynamics: preemption as effective preemption mass

The core idea is that learners update licensing from positive uses of u and from structured non-occurrence when u is a plausible competitor and is repeatedly not chosen. The update shouldn't treat all "opportunities" as equally informative: absence is evidential only to the extent that u would have had non-trivial probability of being chosen if it were licensed.

For a target variant u associated with niche $n(u)$, define the EFFECTIVE PREEMPTION MASS in a time window as the expected number of u-tokens that would have been produced if u were a licensed competitor:

$$p_t(u, c) = \sum_{j=1}^{N_t(n(u),c)} \rho_t^\star\big(u \mid n(u), c\big),$$

where $N_t(n(u), c)$ is the number of niche opportunities observed in that window. This quantity is large only when (i) the opportunity set is large and (ii) u would have been chosen at non-trivial rates if licensed.

A simple Bayesian learner for licensing represents each construction node (g, c) with a Beta posterior $\mathrm{Beta}(a_{g,t}, b_{g,t})$, updated by three evidence streams: s_t (positive evidence), e_t (error evidence), and p_t (effective preemption mass). An observed token or missed opportunity distributes its evidence over activated nodes by the coverage weights $\lambda_g(u, c)$:

$$\begin{aligned} a_{g,t+1} &= a_{g,t} + \lambda_g(u, c) s_t(u, c), \\ b_{g,t+1} &= b_{g,t} + \lambda_g(u, c)\{e_t(u, c) + p_t(u, c)\}. \end{aligned}$$

The implied node-level mean licensing rate is then:

$$C_{t+1}(g, c) = \frac{a_{g,t+1}}{a_{g,t+1} + b_{g,t+1}}. \tag{28}$$

Token-level $C_t(u, c)$ is obtained by the partial-pooling equation in §3.2.

The streams themselves are ascription-weighted. The definition of Λ already excludes performance slips and nonce errors, and the conditioning state already fixes whose norms are being estimated; making this explicit, each token's contribution to s_t, e_t, or p_t is scaled by the hearer's ascription weight: the probability that the token is a competent realization of the norms of c, given cues about the producer (age, learner status, group membership) and about the token (disfluency, self-repair). Child and learner errors barely move adult posteriors even in error-dense environments, not because they go unheard but because they're attributed to development or transfer rather than to the norm-centre. The same machinery as $q_h(c \mid e)$ does the work: a token is evidence about the norms of whatever conditioning state its producer is taken to competently instantiate. Contact-driven change (koinéization) then falls out as a shift in ascription weights rather than a separate mechanism.

The two kinds of evidence also attach at different grains. Positive evidence distributes over activated nodes by the coverage weights: a heard token confirms every construction that covers it. Error and preemption mass instead concentrate on the most specific node that individuates the candidate within its niche, because that node is what distinctively predicts the missing tokens. In Bayesian terms, the likelihood penalty for structured non-occurrence falls on the hypothesis that wastes probability mass on sentence types never observed; likelihood is a quantitative measure of the weight of implicit negative evidence (Perfors et al., 2011).

Partial pooling raises an objection. Token-level licensing is a convex sum (§3.2):

$$C_t(u, c) = \sum_g \lambda_g(u, c)\, C_t(g, c).$$

Left-branch extraction activates the well-licensed nodes for interrogative fronting, question formation, and modifier–noun packaging. The sum can look as though it should hand the candidate substantial licensing.

The coverage weights block that result. A node contributes to a token's pooled licensing only for the properties it actually sanctions, and those parent nodes sanction only *contiguous* realizations. None of them covers the determiner–head discontinuity that's the distinctive property of **Which did you buy car?*; producing that token requires a node that licenses a determiner–head arc crossing the verb, and no contiguity-preserving node in $\mathcal{H}$ generates one (Reynolds, 2026d).

The parents keep their high licensing, each confirmed by its own abundant contiguous tokens, and contribute none of it to the discontinuous token through the pooling sum. The dedicated left-branch-extraction node, the only activated node that covers the discontinuity, carries near-gating weight for it and absorbs the preemption mass from every fronting opportunity where a contiguity-preserving competitor wins. The dedicated node is required by the derivation, not gerrymandered to force the result.

This asymmetry also answers an acquisition worry. If token-level licensing were just pooled parent licensing, learners should pass through a stage of treating left-branch extraction as live. They don't need to: the candidate is generated by analogy, so it's on the table with non-zero prior support, but its dedicated node has no scaffolding of its own, fronting opportunities are dense in the input, and each one adds preemption mass. A companion corpus study quantifies the evidential situation: across 1,228 fronting opportunities in a sixteen-corpus Universal Dependencies sample, zero genuine determiner–head discontinuities occur (95% upper bound 0.24%), while contiguity-preserving alternatives (pied-piping, fused-head construals, and the *big mess* family) occur at measurable rates

(Reynolds, 2026d). Comprehension pushes the same way: a fronted *whose* or *which* defaults to a fused-head parse, so the discontinuous analysis also loses the parsing race (Culicover et al., 2022). The non-satiation of left-branch extraction, confirmed in a meta-analysis of 25 experiments (Lu et al., 2024), is exactly what a concentrated near-zero posterior predicts (§3.3.2).

Decaying-gain learning and logistic dynamics. The discrete update (28) is the primary dynamics. Applied to a fixed environment (exogenous evidence streams), it's a decaying-gain estimator: the posterior mean follows a hyperbolic trajectory of the form $a_0/(a_0 + b_0 + \bar{e}\, t)$, as plotted in Figure 5, and converges without bifurcating.

The logistic approximation $\dot{C} = r(g, c)\, C(1 - C)$ arises when the loop in §3.3.2 is closed: positive evidence is generated by production at a rate that grows with the current licensing state, while error and preemption mass accrue when the target isn't chosen. Replacing those endogenous evidence streams by their expectations yields a logistic drift, with $r(g, c)$ a net evidence rate and a gain that's constant under bounded evidence memory (decaying otherwise). I use this ODE only for qualitative fixed-point and comparative-statics arguments; the fixed points at 0 and 1 belong to the closed-loop population dynamics, not to individual open-loop learning.

4.4 Emergent categoricality as bimodality

The sense of emergence used here is weak and mechanistic. Population-level categoricality is derived from learner-level updating, selection, and repair; it isn't a label for whatever remains unexplained. This differs from Hopper's "emergent grammar", where grammar is perpetually provisional (Hopper, 1987). The present model allows conditioned states with near-absorbing attractors.

In the mean-field approximation, $\dot{C} = rC(1 - C)$ has fixed points at 0 and 1. When $r > 0$, positive evidence dominates and the licensing posterior is driven toward 1. When $r < 0$, error evidence and effective preemption mass dominate and the posterior is driven toward 0. Across construction nodes, a community with many evidence rates bounded away from zero develops a bimodal distribution of C_t: most nodes cluster near licensed or non-licensed attractors, while only low-evidence or contested nodes remain diffuse.

This gives the formal sense in which categorical grammaticality emerges. Closure raises $\rho_t^\star$ by keeping the competitor set small and role-specific; large $N_t(n, c)$ supplies repeated opportunities; their product is the effective preemption mass. Closed, high-opportunity, update-critical contrasts are exactly the parameter regime in which the licensing posterior becomes bimodal. A finite-population stochastic version should strengthen this pattern: sampling drift near 0 and 1 makes the boundaries sticky, while middle states remain vulnerable to renewed evidence. This connects the present dynamics to complex-adaptive and cultural-evolutionary accounts of language change (Beckner et al., 2009; Blythe & Croft, 2012; Kirby et al., 2014), and to invisible-hand explanations, where macro-level order emerges as the unintended consequence of micro-level choices (Keller, 1994).

4.5 Actuation

The same dynamics define actuation: a sign change in the expected evidence balance for licensing. Actuation occurs when the expected positive stream $s_t(u, c)$ begins to outweigh the combined negative streams $e_t(u, c) + p_t(u, c)$ across successive windows, so that the posterior mean $C_{t+1}(u, c)$ in (28) increases rather than decays. In the mean-field approximation from §4.3, the net evidence rate $r(u, c)$ becomes positive, yielding the familiar S-curve at the population level.

The factors that drive such bifurcations align with the motivations discussed in §2.5. Semantic reanalysis may increase utility (and thus $\rho_t^\star$) by making a form–value relation more transparent or useful. Social pressures may shift utilities (prestige effects) or alter the relevant community standard. Structural motivations such as analogical extension can systematically raise licensing probability by borrowing strength from frequent neighbors. Processing innovations may reduce error rates (e_t) or locality costs, improving the net evidence balance.

Individual innovation alone won't do. A few speakers adopting a marginal construction don't guarantee its success; the evidence balance has to shift so that acceptance systematically outweighs rejection across the relevant community. Many sensible innovations fail because they appear before those community conditions are in place.

Near the critical point, small perturbations in community attitudes can trigger rapid shifts between rejection and acceptance. As the balance becomes more strongly positive, the construction moves further from the unstable rejection state, making reversal increasingly unlikely. This produces the characteristic S-curve of documented language changes (Kroch, 1989). For constructions in the marginal zone, with near-zero evidence balance, the model predicts heightened sensitivity to external factors and greater cross-community variation. Small or peripheral communities may show volatile behaviour near bifurcation points before a change spreads to larger population centres.

4.6 Apparent exclusions without independent constraints

When $\mathsf{map}(u, c) = 1$ and $K(u, c)$ is high but $C_t(u, c)$ is driven toward zero by persistent preemption in a large opportunity set, speakers converge on a sharp, non-satiating rejection profile. Additional processing penalties in $\mathrm{R}_i(u, c)$ (e.g. systematic garden-path repair due to an entrenched fused-head construal) can strengthen the subjective categoricality without any independent structural constraint being posited.

5 Theoretical implications

The payoff of OVMG isn't just a new decomposition of grammaticality. It says what the category should let us expect. If a form–value relation is grammatical in a communicative situation, it should pattern with other licensed resources in repair behaviour, transmission, satiation under exposure, and trajectories of change. If it is unlicensed, weakly licensed, or backed by little evidence, those expectations should fail in different, diagnostic ways (§3.3.2).

This is the sense in which the account is projectibility-first: grammaticality earns theoretical standing by the inferences it supports, not by the mere fact that a mechanism can be named for it (Boyd, 1999; Goodman, 1955; Khalidi, 2013, 2018; Millikan, 1999; Reynolds, 2026a; Slater, 2015). The profile that supports those inferences is conditioned form–value stability. The evidence offered here reaches two levels. First, bimodal licensing dynamics explain why some form–value relations stabilize near acceptance or rejection. Second, the production–repair loop explains how later use can help maintain that state. A stronger claim would require evidence for corrective control: a process that registers departures from the licensed profile and pushes the community back toward it. That stronger claim is left open (Reynolds, 2026b; Woodward, 2001, 2003).

This recasts the competence–performance relation. Licensing and feeling aren't two fundamentally different phenomena but different manifestations of the same conditioned state: processing and other performance factors help shape which form–value relations stabilize, and those stable relations in turn constrain performance. The identification strategy in §3.6 keeps this from becoming an escape hatch, since licensing and feeling are measured by different evidence streams and can falsify each other. The rest of this section develops the consequences: the account's reading of macro-typological universals, its distinctive predictions, and its relation to neighbouring frameworks.

5.1 Macro-typological constraints on grammatical design

Recent macro-typological work sharpens the question of how strongly grammar is constrained across languages. Verkerk et al. (2025) test 191 implicational universals from the Universals Archive against Grambank's 2,430-language morphosyntactic sample, using Bayesian models that control explicitly for genealogical and areal non-independence and then follow up with spatiophylogenetic analyses of evolutionary rates.

A naïve analysis that ignores relatedness appears to support the vast majority of proposed universals, but once phylogeny and geography are accounted for, only about a third (60 of 191) remain statistically supported. Support is concentrated in relatively narrow domains: most hierarchical universals and a substantial minority of "narrow" word-order universals survive, whereas "broad" word-order universals and miscellaneous others fare poorly. Diachronically, supported universals typically correspond to "harmonic" combinations of features (for instance, consistent head–dependent orders) that languages are more likely to evolve into than out of, with these preferred configurations recurring independently across lineages.

From an OVMG perspective, these findings identify candidate attractors in the design space of operator-valued form–value relations rather than exceptionless grammatical laws. They don't by themselves show that a macro-typological feature is a language-internal grammatical object. The typological claim has to pass through a comparandum-to-realization mapping: one must specify which comparative feature is being tracked, how each language realizes the relevant operator value or form–value relation, and whether the resulting profile predicts held-out languages or diachronic trajectories (Reynolds, 2025).

Under that discipline, supported patterns correspond to configurations for which the net evidence balance (positive evidence s_t vs. preemption/error mass $e_t + p_t$) in the dynamics of §4.3 is positive across a wide range of communities. They're cognitively and communicatively favourable enough that repeated episodes of change tend to push $C_t(u, c)$ toward fixation in lineage after lineage.

The fact that roughly two-thirds of the tested universals fail once autocorrelation is controlled for also fits a de-idealized view of grammaticality. Community-specific conventions still have considerable freedom in how they realize operator values, with only some regions of the space strongly preferred. Large-scale comparative work of this kind complements OVMG by locating candidate stable regions; the present framework explains how local community dynamics and form–value coherence can make those regions reachable and persistent.

5.2 Predictions

The model's distinctive predictions come from separating positive evidence, opportunity structure, and counterfactual choice probability. Cross-linguistic gender remains a useful boundary case, but it isn't the strongest test. Any framework that treats grammaticalized concord as obligatory predicts stronger judgments where concord is more tightly grammaticalized. The companion operator-stratum account also complicates the simple gender story: gender concord can be policed through paradigm inertia even when its public-update value is attenuated (Reynolds, 2026c). The sharper predictions concern preempted gaps, artificial learning, and L2 transfer.

SATIATION FRAMING. Repeated exposure should attenuate negative feelings most strongly when a form is low-evidence rather than strongly preempted. Independent relative *whose*, as in ?**The packages are still here, but Nathan, whose was open, just left.*, should be sensitive to framing: participants told that tokens come from an intentional dialectal, literary, or ingroup resource should show more improvement than participants told that the same tokens are errors. Left-branch extraction should resist the same treatment because its opportunity set is large and its effective preemption mass is high. The dependent variables are rating change, later repair behaviour, and elicited production.

ARTIFICIAL-LANGUAGE LEARNING. An artificial-language-learning experiment can hold positive evidence constant at zero while manipulating opportunity-set size. Learners could see a system in which a target variant is never produced. In the high-opportunity condition, competitor variants repeatedly occupy the relevant niche and the target has high $\rho_t^\star$; in the low-opportunity condition, the niche rarely arises or the target would have low counterfactual utility. OVMG predicts categorical rejection in the first condition and uncertainty in the second, even though both conditions contain identical zero positive evidence. The design operationalizes the Figure 5 contrast and is compatible with artificial-language learning paradigms that manipulate simplicity and communicative structure (Culbertson & Kirby, 2016).

L2 PREEMPTION TRANSFER. Early L2 rejections should track L1 preemption mass, not just L2 frequency. A learner whose L1 strongly preempts a candidate in a high-opportunity niche should reject the L2 analogue early, even if the L2 input frequency is low enough to leave native speakers uncertain. Conversely, low L1 preemption should make learners more willing to treat rare L2 patterns as unresolved rather than impossible. As learners join the L2 speech community, $q_h(c \mid e)$ and the constructional hierarchy should shift toward the target norm-centre. This view aligns with Selinker's (1972) concept of INTERLANGUAGE, but gives a more specific prediction: transfer is strongest where the L1 has accumulated large effective preemption mass.

INTONATIONAL OPERATORS. Nothing in the model privileges segmental material, so intonational patterns are predicted to count as operator exponents wherever they form closed, high-opportunity, update-critical repertoires: the English polar-question rise and nuclear-accent placement configure what an utterance does to the common ground much as tense and agreement do (Reynolds, 2026c). The Turkish suffix-harmony analysis in Appendix A gives the parallel case for segmental exponent choice: phonological material matters for grammaticality when it realizes an operator exponent. The corresponding intonational test is whether mismatches between intonation and text (e.g. nuclear accent on a closed-class word under broad focus) are policed like grammar, with categorical rejection, repair, and resistance to framing, or like style.

5.3 RELATIONSHIP TO GENERATIVE GRAMMAR

Generative work made a central contribution by showing that grammaticality can't be reduced to semantic plausibility, processing ease, or frequency of attestation. Chomsky's *Colorless green ideas sleep furiously* made the point vivid: speakers can recognize syntactically well-formed sentences even when they're semantically bizarre. Generative analyses explain categorical constraints like the impossibility of extracting determiner-adjective sequences in English (**Which do you prefer car?*), and they capture the fact that such sequences remain ungrammatical even when their intended meaning is clear and processing demands are low.

OVMG keeps that lesson. Grammaticality judgments reflect systematic patterns (Dennett, 1991); those patterns can't be reduced to meaning or processing alone; and certain syntactic configurations appear to be categorically excluded regardless of context. The earlier diagnostics for center embeddings and preempted gaps build directly on ideas about recursive structure and acknowledge the generative discovery that some constructions appear to be excluded by grammar-internal constraints.

The same point governs the present account's use of formal evidence. A generative analysis can supply a real warrant when it specifies the target, diagnostics, evidence, and failure conditions for a grammatical contrast. OVMG isn't an attempt to replace every I-LANGUAGE analysis; it rejects only the stronger claim that categorical grammaticality must be explained by an autonomous syntactic component or a UG-backed inventory of primitives. Its target is the community-level licensing profile that makes some form–value relations categorically policed.

One well-known case is independent relative *whose* (1d). As Hankamer and Postal (1973) observe, the construction appears to violate no syntactic principles: independent genitives are possible (*Mine was visiting*), independent interrogative *whose* is grammatical (*Whose was open?*), and *whose* can be used as a dependent relative pronoun (*the student whose friend was visiting*). The generative tradition's careful documentation of such cases, where seemingly parallel constructions show puzzlingly different grammatical status, has driven theoretical development.

OVMG explains the contrast differently. Rather than positing an autonomous syntactic component, it treats grammatical constraints as products of form–value relations within specific communicative situations. For independent relative *whose* to be felicitous, multiple conditions have to converge: the possessor has to be sufficiently accessible in the discourse while the possessum is predictable enough to license ellipsis, but the possessive relationship needs to be semantically significant enough to warrant explicit marking, and this configuration has to occur in a context where a relative clause is the natural way to package this information. The extreme rarity of contexts satisfying all these conditions appears to prevent the construction from becoming conventionalized as an English resource: speakers encounter it so rarely, despite perfectly common components, that even when all conditions align perfectly, the construction feels alien.

The difference is evidential rather than merely terminological. A generative theory has to explain why a syntactically possible and pragmatically useful construction is systematically avoided. OVMG treats extreme mismatches between predicted and observed frequency as possible evidence of strong negative weighting, even when the exact source of that evidence remains unclear. The result preserves the generative recognition of systematic constraints while embedding it in a theory of how form–value relations become established and maintained in language communities.

Other cases that generative grammar struggles to explain are grammatical with one meaning but ungrammatical with another, such as (13) *I have 16 years*. While syntactically identical to grammatical expressions like *I have 16 dollars*, this construction becomes ungrammatical specifically when used to express age. A purely syntactic account would have to explain why the same structure is well-formed in one case but ill-formed in another, despite no apparent syntactic differences.

OVMG, in contrast, locates the source of ungrammaticality in the community's form–value conventions: *have*+numeral years has become conventionalized for expressing duration or future time (*I have 16 years until retirement*) but lacks positive evidence for expressing age, where a different construction (*I am 16 years old*) is the established pattern. Similar cases arise with plural forms that are grammatical with some meanings but not others (e.g. *peoples* for ethnic groups but not multiple individuals) and with verbs that resist certain arguments despite no obvious syntactic prohibition (e.g. *discuss about*). These meaning-dependent grammaticality patterns suggest that what gets down-weighted or up-weighted often depends on specific form–value associations rather than purely structural constraints.

5.4 Relationship to Construction Grammar

Construction Grammar (CxG) (Fillmore, 1988; Fillmore et al., 1988; Goldberg, 1995, 2019; Kay & Fillmore, 1999; Sag, 2012) made a central contribution by treating learned form–meaning pairings as grammatical objects. At its core, CxG argues that language consists of learned relationships between form and meaning at multiple levels of complexity. These form–value relations, or constructions, include both individual morphemes and abstract syntactic patterns. This perspective helps explain phenomena that proved challenging for earlier approaches, including idiomatic and partially schematic expressions that don't fit a clean core/periphery split.

CxG shows that meaning suffuses all levels of grammatical organization. Rather than treating syntax as an autonomous formal system that interfaces with semantics only at designated points, CxG reveals how meaning and form are inseparable aspects of linguistic knowledge. For instance, the *What's X doing Y?* construction carries an implication of incongruity that can't be derived from its component parts (Kay & Fillmore, 1999). Recent work connects this framework to model-internal analysis: Weissweiler et al. (2023) use construction grammar to probe how neural language models handle different levels of linguistic abstraction.

OVMG shares these CxG commitments about form–value relations and constructional abstraction. It doesn't privilege one expressive substance over another. The companion operator-stratum paper explicitly denies a substance-based boundary: phonological, gestural, lexical, morphological, and syntactic material can all realize operator contrasts when they enter a closed, accountable public-update repertoire (Reynolds, 2026c). The OVMG claim is narrower. Categorical grammaticality clusters around form–value relations with a particular role: they're high-opportunity, closed-paradigm contrasts that configure update, role allocation, scope, reference, or repair.

The contrast becomes clear when speakers judge violations. Register, politeness, genre, and many lexical choices can be stable and socially important without blocking the public update itself. A wrong tense, agreement value, clause-linking marker, or obligatory allomorph can mis-set the update instruction even when the intended message is recoverable. The age-predication example (*I have 25 years*) is transparent but not licensed for that English frame; its rejection comes from non-licensing in a high-opportunity constructional niche, not from the substance of syntax or morphology as such.

OVMG preserves CxG's systematic treatment of constructions while adding a prediction about which constructional contrasts attract categorical policing. The relevant contrasts are those whose closure and opportunity structure drive the licensing posterior toward the bimodal regimes in §4.4. That lets CxG's continuum of constructions coexist with a non-arbitrary explanation of why some violations are heard as grammar and others as style, stance, or appropriateness.

5.5 Relationship to usage-based approaches

OVMG shares with usage-based approaches (UBA) (Bybee, 2006, 2007, 2010) the idea that linguistic knowledge develops from patterns of actual language use rather than from an autonomous formal system. Both perspectives reject the notion that grammaticality can be reduced to abstract rules operating independently of meaning and context.

Several pieces of the formal core come directly from this tradition. The individual/population split ($\Lambda_{i,t}$ for a speaker's own licensing state, C_t for the population rate) restates Schmid's (2020) separation of ENTRENCHMENT (the cognitive reorganization of a speaker's knowledge) from CONVENTIONALIZATION (the social establishment of a community's regularities), the two coupled feedback cycles of his Entrenchment-and-Conventionalization model. The population dynamics themselves descend from Croft's (2000) utterance-selection theory, in which variants are replicated and selected within a community; the S-curve results the model reuses are already worked out in that lineage (Blythe & Croft, 2012).

Frequency per opportunity is likewise not new: it's the core of Stefanowitsch's (2006) case that corpora contain negative evidence, where a form counts as "significantly absent" rather than "accidentally absent" only once its observed frequency is weighed against the frequency expected given its opportunities. Variationist sociolinguistics has normalized by opportunity since Labov's principle of accountability (Labov, 1972).

OVMG adds their joint formalization and a prediction. The licensing-versus-selection factorization $\pi_t \approx C_t \cdot \rho_t^\star$ turns Croft's replication-versus-selection into an estimable product; effective preemption mass $p_t = N_t \cdot \rho_t^\star$ puts a quantity on the absence that Stefanowitsch left open, since he showed how to establish that a form is significantly absent but stressed that significant absence "does not, in itself, provide any clues as to why". Licensing-versus-selection is exactly that missing clue: a significantly absent form may be unlicensed ($C_t \approx 0$) or licensed but dispreferred ($\rho_t^\star \approx 0$), and the two project differently. The Beta dynamics then convert these quantities into the bimodality result (§4.4) and the prediction about which contrasts attract categorical policing. The relation to Goldberg's (2019) COVERAGE and statistical preemption is one of formalization in the same spirit: her "explain me this" puzzle (a high-coverage double-object frame preempted by the prepositional dative) is a preemption-mass story in which the competitor wins a high-opportunity niche. One caution follows from Ambridge et al. (2015), who find that overall form frequency (entrenchment) predicts the retreat from overgeneralization more robustly than competitor frequency (preemption), and who propose collapsing the two into a single construction-competition process. OVMG keeps them analytically separate: error evidence e_t and preemption mass p_t are distinct streams, but they enter the Beta update additively (§4.3), so the model can represent Ambridge and colleagues' single-process view as their sum while leaving the separability an empirical question rather than a stipulation.

The analysis of independent relative *whose*, as in ?*I saw Joan, a friend of whose was visiting*, is a case in point. A simple UBA account might predict that this construction's marginality stems from its low frequency. But this explanation proves insufficient: the construction is rare, and it's dramatically rarer than one would expect given the frequency of its component parts. Independent *whose* appears in interrogatives (*Whose is that?*), and the relative *whose* is common in dependent contexts (*the student whose paper was late*). Given these frequencies, analogical extension should make the independent relative more common than observed.

Raw component frequency is therefore the wrong predictor; *whose* isn't a preempted gap. The frequency of independent and relative *whose* taken separately doesn't create opportunities for the niche that requires both at once, together with an accessible possessor, a predictable possessum, and a licensing ellipsis site; that convergence is vanishingly rare. This isn't the significant absence of a preempted form, which needs a large opportunity set and a competitor that keeps winning; it's the low-opportunity uncertainty described in the diagnostic categories above. The tension between "rarer than expected" and "uncertain rather than rejected" dissolves once licensing is separated from selection: *whose* is under-*observed* because its niche rarely arises, not preempted because a rival repeatedly beats it.

OVMG doesn't exempt the operator stratum from usage dynamics; it makes those dynamics more specific. A rare form can remain grammatical when the opportunity set is tiny or when higher-grain constructions license the token by partial pooling. A transparent candidate can become categorically rejected when the opportunity set is large, the analogical candidate would have had high $\rho_t^\star$, and an entrenched competitor wins every time. The result preserves usage-based findings about frequency and entrenchment while predicting when frequency becomes preemption.

5.6 Relationship to logicality of language accounts

An alternative perspective, prominent in recent formal semantics, suggests certain types of unacceptability stem from the language faculty itself possessing a deductive system that identifies and filters sentences with logically trivial meanings (tautologies or contradictions) (cf. Chierchia, 2013; Del Pinal, 2019; Fox, 2000). This "logicality of language" hypothesis attempts to explain, for example, systematic restrictions on quantifiers by arguing the unacceptable cases are "L-trivial", meaning their triviality arises solely from the meaning and configuration of logical or closed-class terms (like *every*, *some*, *not*), irrespective of the open-class words (like *student*, *run*) (Gajewski, 2009). A key challenge is explaining why simple tautologies or contradictions (e.g. *It's raining and it isn't raining*) are often acceptable. Del Pinal (2019) argues against "Logical Skeletons" (which assume the system ignores open-class word identity) in favour of "LF+RESCALE", a view where the system sees standard logical forms but allows optional, context-dependent modulation of open-class terms (e.g. interpreting the second "raining" as "raining hard") to yield non-trivial meanings.

OVMG offers a broader account of ungrammaticality. "Logicality" approaches explain restrictions tied to logical and closed-class vocabulary through L-triviality. OVMG also covers cases less easily reducible to logical contradiction, such as absent form–value relations (1a), strong deviations from conventional community forms (17), and extreme, unexpected rarity (1d). By grounding grammaticality in community-specific conventions and allowing for gradient compatibility ($K(u, c)$) and acceptance ($C_t(u, c)$), OVMG accommodates cross-linguistic variation and degrees of acceptability, which are less central to the L-triviality filter.

LF+RESCALE provides a specific mechanism for acceptable "trivialities". OVMG treats those cases as possible products of more general interpretive principles within community norms, without positing a dedicated deductive module or specific operators like RESCALE. It also keeps objective grammatical status separate from subjective processing effects or "feelings" (§3.7). OVMG frames grammaticality as an emergent consequence of communicative practice in the formal sense of §4.4, rather than a direct output of logical computation within syntax.

5.7 Relationship to relevance-theoretic accounts

Recent work by Scott-Phillips (2026) makes a useful observation: linguistic intuitions about acceptability arise as byproduct effects of cognitive systems for interpreting communicative acts. Just as people immediately sense when a visual stimulus violates core assumptions about physical objects (as with impossible objects), they detect when utterances violate basic presumptions about communicative efficiency. Scott-Phillips argues that unacceptability occurs not from mere inefficiency, but from

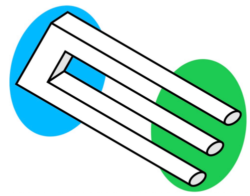

Figure 6: Impossible trident

an inherent impossibility of interpreting an utterance consistently with these presumptions, much as an impossible trident (Figure 6) can't be interpreted as physically cohesive in any context.

OVMG shares several premises with this account. Both reject the need for an innate grammar faculty, locating linguistic intuitions instead within general cognitive systems, and both treat language as developing from communicative needs rather than autonomous syntactic principles. OVMG's emphasis on situation-specific form–value relations builds directly on Scott-Phillips's arguments about how communicative pressures shape linguistic conventions.

The frameworks differ primarily in their explanatory mechanisms. Where Scott-Phillips argues that grammaticality judgments reduce to impossibilities of efficient interpretation, OVMG suggests that while communicative pressures shape which form–value relations become conventionalized, these relationships then create systematic constraints that can't be reduced to efficiency alone. For Scott-Phillips, one would have to demonstrate that (1c) contains inherent contradictions making efficient interpretation impossible. OVMG instead analyzes how the present perfect's tense–aspect operator value clashes with the meaning of the lexeme *yesterday*. While these analyses might ultimately converge, it remains unclear why the criterion of inherent impossibility of interpretation should apply specifically to operator-value violations rather than to lexical-lexical conflicts or certain phonological patterns. The scope of what constitutes an interpretive impossibility requires further theoretical development.

Empirically, OVMG requires tests of whether specific form–value relations are stable within a communicative situation and where operator-value conflicts arise. The challenge for relevance-theoretic accounts, as Scott-Phillips (personal communication, Dec. 16, 2024) acknowledges, lies in establishing independent, empirically vulnerable claims about what makes efficient interpretation inherently impossible rather than merely difficult. The clearest comparison would test the two accounts on novel constructions as they become acceptable or unacceptable.

6 Limitations and future directions

The account has real limits, and each points to work it invites rather than forecloses. One is its reliance on the concept of "communicative situation". Speakers move among overlapping contexts that shift with the immediate setting, durable social positions, and the norms they orient to. The account builds in this fluidity, but operationalizing these dimensions, and measuring their influence on grammatical stability, will take more precise methods.

A second boundary concerns stylistic variation and individual preference. Most choices that fall within grammatical acceptability are not cases of (un)grammaticality at all. They matter for this account only when they become tied to communicative situations strongly enough to affect licensing, repair, or categorical rejection. The account still has to explain how stylistic preferences become grammatical licensing facts.

A third concerns purely formal constraints that seem to exist independently of meaning or communicative role. The paper argues that many apparently arbitrary restrictions follow from historical processes and competing motivations, but there may be residual cases that resist even that treatment, and they'd sharpen the boundary of the account.

6.1 Methodological considerations

The development of experimental syntax since the 2000s (Schütze, 2016) has transformed how gradient acceptability is measured, letting researchers quantify subtle differences in judgments and track how they change with exposure. The framework's predictions invite corpus-based, experimental, and cross-linguistic investigation. Several avenues stand out:

1. *Corpus analysis:* Large, balanced corpora allow researchers to track frequency patterns and stability over time. Investigating rare or marginal forms (e.g. the independent relative *whose*) in corpora can reveal whether low frequency corresponds to genuinely unstable operator-value patterns or merely a sampling artifact. Comparative corpus research in multiple languages can test predictions about how community values influence grammatical distinctions.
2. *Grammaticality judgment experiments:* Controlled psycholinguistic tasks, including magnitude estimation or forced-choice paradigms, can assess subtle gradience in grammaticality judgments. By manipulating exposure and increasing familiarity with marginal forms, researchers can measure satiation effects and distinguish between entrenched categorical constraints and constructions that become more acceptable through repeated exposure.
3. *Processing measures:* Eye-tracking, self-paced reading, and EEG studies can identify whether some judgments arise from processing overload rather than durable grammatical constraints. If a form becomes easier to parse with practice, it suggests that processing complexity, rather than an immutable rule, lies behind initial judgments of ungrammaticality.
4. *Sociolinguistic fieldwork:* Investigations in communities with distinct dialects or multilingual practices can clarify how social and pragmatic motivations shape grammatical stability. Elicitation tasks and careful comparisons across speech communities can confirm that what counts as grammatical depends on local norms rather than universal principles.
5. *Longitudinal and historical studies:* Diachronic corpora and historical grammars can trace how marginal constructions evolve, testing predictions about which factors lead certain patterns to stabilize, which fade, and which undergo reanalysis. Tracking a community's acceptance of previously dubious constructions over time provides direct evidence for the gradual entrenchment processes posited by the framework.
6. *A threshold-discontinuity test:* The categorical predicate G_t earns standing only if community treatment of a form changes *discontinuously* as the graded state crosses $\tau(c)$, over and above the smooth effect of $\widetilde{G}_t$. That's a regression-discontinuity question. Hearer responses are naturally multinomial (accept, repair, request clarification, sanction), and covariate-adjusted local-likelihood estimators for discontinuities in multinomial response probabilities exist off the shelf (Lazzaro, 2026), with the processing-cost components of R_i entering as predetermined covariates. Two caveats are design constraints rather than afterthoughts: in a confirmatory test, $\tau(c)$ is latent, so it has to be estimated and registered before the outcome data is touched; and the bimodality result (§4.4) predicts that the density of forms near the threshold is thin, so power is scarce exactly where the test bites. The design also yields a further distinctive prediction: because $\tau(c)$ is decision-

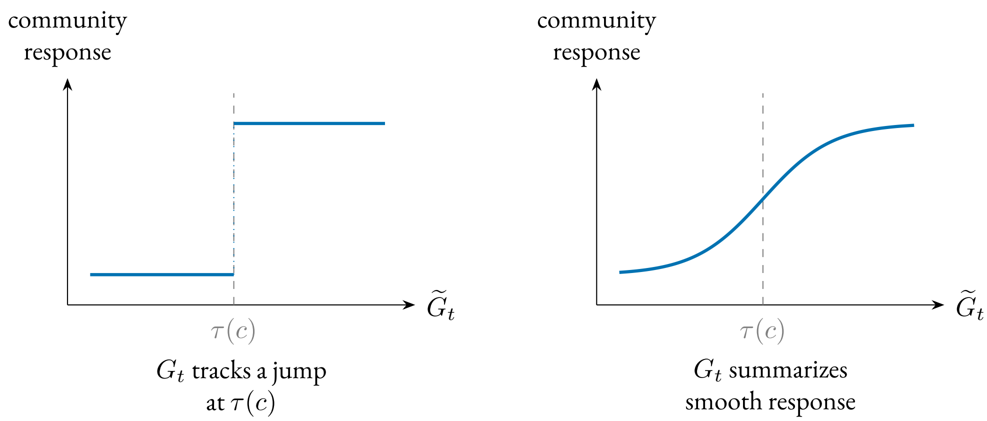

Figure 7: The threshold-discontinuity test. If categorical grammatical status tracks a discontinuity in the population state, community treatment of a form should change discontinuously as $\widetilde{G}_t$ crosses the situational threshold $\tau(c)$ (left); if the predicate only summarizes the graded score, treatment should vary smoothly through it (right). A covariate-adjusted regression-discontinuity design on multinomial hearer responses (Lazzaro, 2026) distinguishes the two.

theoretic (§3.3), independently manipulating the stakes should move the location of the discontinuity. Smoothness through the threshold, by contrast, would leave G_t as bookkeeping over $\widetilde{G}_t$.

Figure 7 states the test in one picture: the categorical predicate tracks a discontinuity only if community treatment jumps at $\tau(c)$.

7 Conclusion

Grammaticality, de-idealized, isn't a categorical fact about an ideal speaker in a homogeneous community but a conditioned form–value relation. Its profile, whether an operator value is structurally realizable (map), coherent (K), and licensed across the population (C_t), is specified relative to a communicative situation and read through a graded licensing posterior. The old competence–performance split becomes two layers with separate evidence streams, a conventional, population-level status and a subjective feeling of anomaly, identified by different measures so that they can falsify each other (§3.6). Categorical grammaticality is then derived rather than stipulated, an emergent bimodality of the licensing state under preemption (§4.4), which leaves independent structural impossibility as a residual case rather than the default explanation.

What the profile is for is projection. A category earns its standing by supporting expectations licensed by partial observation, not by the mere presence of mechanisms: repair, transmission fidelity, satiation, and trajectories of change, over the licensing state as such, its concentration as well as its mean (§3.3.2). The supporting claims are graded separately. Bimodal dynamics stabilize the profile, and the production–repair loop maintains it. The paper doesn't show a corrective system that detects deviations and restores the profile; that stronger claim stays open. On this account, grammaticality matters because it predicts which forms will be repaired, transmitted, softened by exposure, or resistant to change.

These commitments expose the framework to failure. The account predicts that satiation should be more readily inducible for low-evidence constructions than for high-opportunity preempted gaps, that artificial-language learners should distinguish zero evidence from strong preemption, and that early second-language rejections should track first-language preemption mass rather than second-language frequency alone. Most sharply, community treatment of a form should change discontinuously as licensing crosses the situational threshold; otherwise, the categorical predicate adds no explanatory content beyond the graded score (§6.1).

The account doesn't have the simplicity of a one-variable theory. It doesn't reduce grammaticality to a single scalar, a competence–performance split, or a social-pragmatic explanation. Its economy is architectural: the same recurring distinctions (mapping, coherence, licensing, subjective feeling, opportunity structure, and diachronic support) handle cases that otherwise invite separate stories. That economy has to be earned empirically. The variables need independent indicators, and the projections need to fail when those indicators are wrong.

It also carries a significant memory burden. Speakers need not store every utterance as a separate item, but the account asks them to retain many situation-indexed pairings among forms, operator values, constructional neighbours, opportunity estimates, competitors, and repair histories. That is a large, schematized inventory, not a small rule list with exceptions attached (Bybee, 2006, 2010; Goldberg, 2019). If linguistic memory is assumed small in advance, stored generalizations get forced into a rule-plus-exception format, and conditioned licensing is misdescribed as residue.

The program these predictions define is measurement-first. Estimate C_t from converging non-rating indicators (corpus rate per opportunity, elicited production, repair), fix the conditioning state before predicting rather than after, and prioritize cross-linguistic contrasts whose operator status differs, since that's where the account predicts categoricality should form or dissolve. De-idealizing grammaticality trades a tractable idealization for a profile that answers to evidence. It buys a category that predicts: which rejections soften, which forms transmit, and which absences hold.

A Turkish vowel harmony and operator-exponent realization

Turkish illustrates a sharp distinction between LEXICAL disharmony inside stems and ALLOMORPHIC harmony on inflectional suffixes.[3] Only suffixal harmony matters for OVMG, and even there the role is indirect. Vowel backness is not a public-update value; plural and past are. In Turkish, the exponents of those operators must satisfy suffixal harmony.

STEM-INTERNAL VOWELS: DISHARMONY TOLERATED. Loanwords such as *doktor* 'doctor' violate backness harmony, but are fully acceptable; speakers store the form, and the community-level licensing state remains at $C_t(u,c) \approx 1$ and $K(u,c) \approx 1$.

(29) doktor
doctor
'doctor' (disharmonic stem, grammatical)

Because the word contains no malformed operator exponent, the form–value mapping succeeds and

$$\widetilde{G}_t(u,c) = C_t(u,c) \cdot K(u,c) \cdot \mathsf{map}(u,c) \approx 1.$$

SUFFIXAL HARMONY: OPERATOR-EXPONENT REQUIREMENT. Inflectional morphemes are lexicalized with an underspecified vowel; the correct allomorph has to copy [$\pm$BACK] (and, for some suffixes, [$\pm$ROUND]) from the final stem vowel. Using the "wrong" vowel leaves part of the exponent-selection requirement unsatisfied, so the null outcome in the K calculation wins and the word is ungrammatical.

(30) a. kitap-lar
book-PL.BACK
'books' (harmonic, grammatical)

b. * kitap-ler
book-PL.FRONT
intended 'books' (suffix harmony violation)

(31) a. gül-dü
laugh-PST.FRONT.ROUND
's/he laughed' (harmonic, grammatical)

b. * gül-du
laugh-PST.BACK.ROUND
intended 's/he laughed' (suffix harmony violation)

OVMG ACCOUNT. For the ill-formed **kitap-ler* and **gül-du*:

- *Mapping* succeeds: the plural / past node selects an exponent.
- *Compatibility* $K(u,c) \approx 0$: the chosen allomorph fails the harmony condition attached to the exponent, so the plural or past operator lacks a well-formed realization for that stem.

[3] See (Kabak & Vogel, 2001; Sezer, 1981) for discussion of the phonology and (Arık, 2015) for experimental evidence on native judgments.

– *Situational licensing* $C_t(g, c) \approx 1$ at the node level: the plural and past nodes are licensed for every speaker. What's entrenched is the node, not the ill-formed token; node-level licensing pools to the token by the coverage equation in §3.2, and the failure lives in K.

So $G_t(u, c) = 0$ and speakers judge the word as categorically wrong, not just odd-sounding. By contrast, *doktor* in (29) keeps $K(u, c) = 1$; harmony is a phonotactic preference that affects only the phonological well-formedness component of $F_{i,t}(u, c)$, not grammaticality.

Localized exceptions. Some derivational suffixes (e.g. *-imsi* '-ish') are lexically marked "disharmonic". Speakers store them, and community-level licensing remains at $C_t(u, c) \approx 1$: no exponent condition is violated, so $G_t(u, c) = 1$ despite vowel mismatch. Clitic boundaries that start a fresh harmony cycle (Kabak & Vogel, 2001) are handled the same way: they satisfy the operator-exponent requirement and leave harmony to phonology alone.

Turkish suffix harmony errors are therefore grammaticality failures only in this indirect sense. They do not mis-set a public-update contrast themselves, but they prevent a closed inflectional operator from receiving a well-formed exponent. Stem disharmony lowers phonological well-formedness and affects only the listener's feeling $F_{i,t}(u, c)$.

A counterexample would be a case where phonology alone produced feelings of ungrammaticality without involving an operator contrast or an operator-exponent requirement.

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